

# EXPLORING THE STARS AND THE MILKY WAY

spacing  
is uneven

move up to be in line  
with the other side

SPONSORED BY:



SWIN  
BUR  
NE

SWINBURNE  
UNIVERSITY OF  
TECHNOLOGY

SPONSORED BY:



SWIN  
BUR  
NE

SWINBURNE  
UNIVERSITY OF  
TECHNOLOGY



## Credits:

This booklet was created by the ASTRAL 2024 and 2025 teams, and their mentors. This booklet was supported by the ARC Centre of Excellence for Gravitational Wave Discovery (OzGrav).

### ASTRAL 2024 COHORT

- Hendrik Combrinck
- Rebecca Koehne
- Ruoyao Ni
- Rudra Sekhri
- Evie Spilias

### MENTORS:

- Professor Matthew Bailes
- Dr Kirsten Banks
- Dr Emma Carli
- Dr Chris Flynn
- Diana Haikal
- Akhil Jaini
- Saurav Mishra
- Bailee Wolfe

### ASTRAL 2025 COHORT:

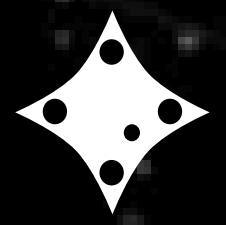
- Lead Editor: Annabelle Leung
- Agatha Beare
- Aaron Choi
- Sevryn Danilov
- Rhea Fernandez
- Lucy O'Shea
- Henryk Przychodzen
- Daniel Rosina
- Tristan Than
- Evie Vale
- Mandini Vyas
- Kim Yan

### SPONSORED BY:



SWINBURNE  
UNIVERSITY OF  
TECHNOLOGY





ASTRAL

if this will  
be kept needs  
to be centered

# EXPLORING THE STARS AND THE MILKY WAY

~~\* Rough Draft~~ (feels like we need more) ~~of stars~~  
Our sun is one star amongst millions. Like people, they  
come in ~~all~~ different colours and sizes. When these  
celestial bodies die, they do so exceptionally, either  
exploding ~~spectacularly~~ or fading quietly. Most are surrounded  
by ~~either~~ planets or <sup>remain</sup> in ~~an~~ orbit with another star. Stars are  
all grouped together in a system, spanning across  
billions of kilometres.



# Introduction

note:  
→ hard to understand  
→ unclear grammar  
→ writing could be more advanced.

~~We~~ humans live on a planet called Earth, ~~which you~~  
~~might already know~~ is orbiting our Sun. But the Sun is  
just one star among millions, stars of all different sizes  
and colours. Stars which explode spectacularly or fade  
quietly, surrounded by planets, or in a system with  
another star. Stars which are all grouped together in an  
object billions of billions of kilometres across.

We call this ~~object~~ <sup>system</sup> the Milky Way.

This booklet will take you on a journey through it,  
seeing all the different types of stars it is home to  
along the way.

## About ASTRAL

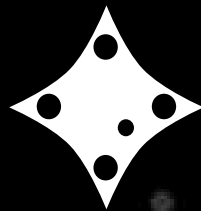
→ repetitive words  
i.e. founded, stars

ASTRAL stands for Astrophysics, Supercomputing,  
Technology, Research, Analysis and Leadership. It was  
founded by Professor Matthew Bailes who also founded  
the Centre for Astrophysics and Supercomputing at  
Swinburne University of Technology. The program was  
founded to inspire students to explore their interests in  
these areas of studies through workshops, programs,  
and outreach projects (~~like writing this booklet~~). By  
providing students with mentors and unique  
opportunities, ASTRAL aims to build a cohort of STEM  
student leaders.

→ any word in  
between  
→ savvy  
→ talented



*this page  
should come  
before the  
intro*



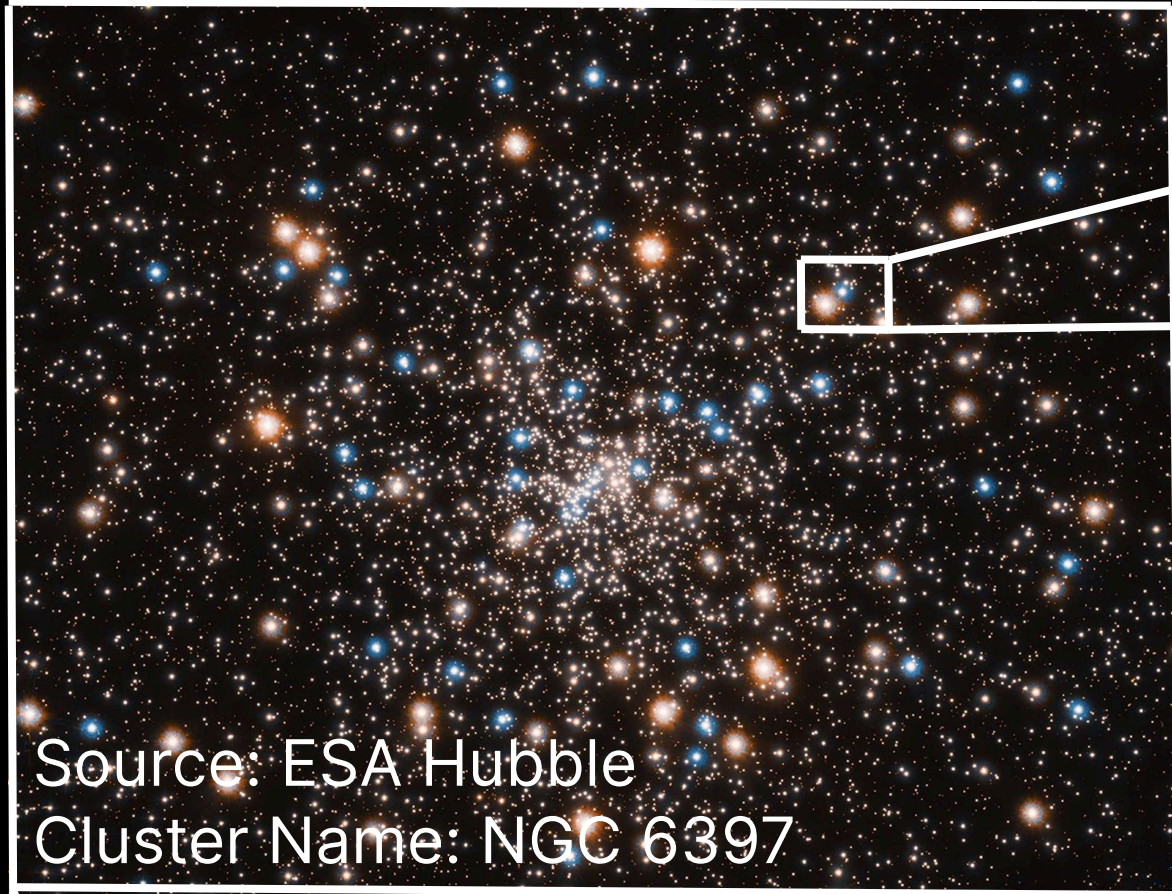
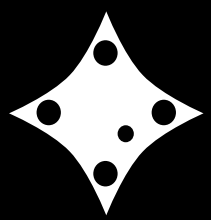
ASTRAL

# Table of Contents

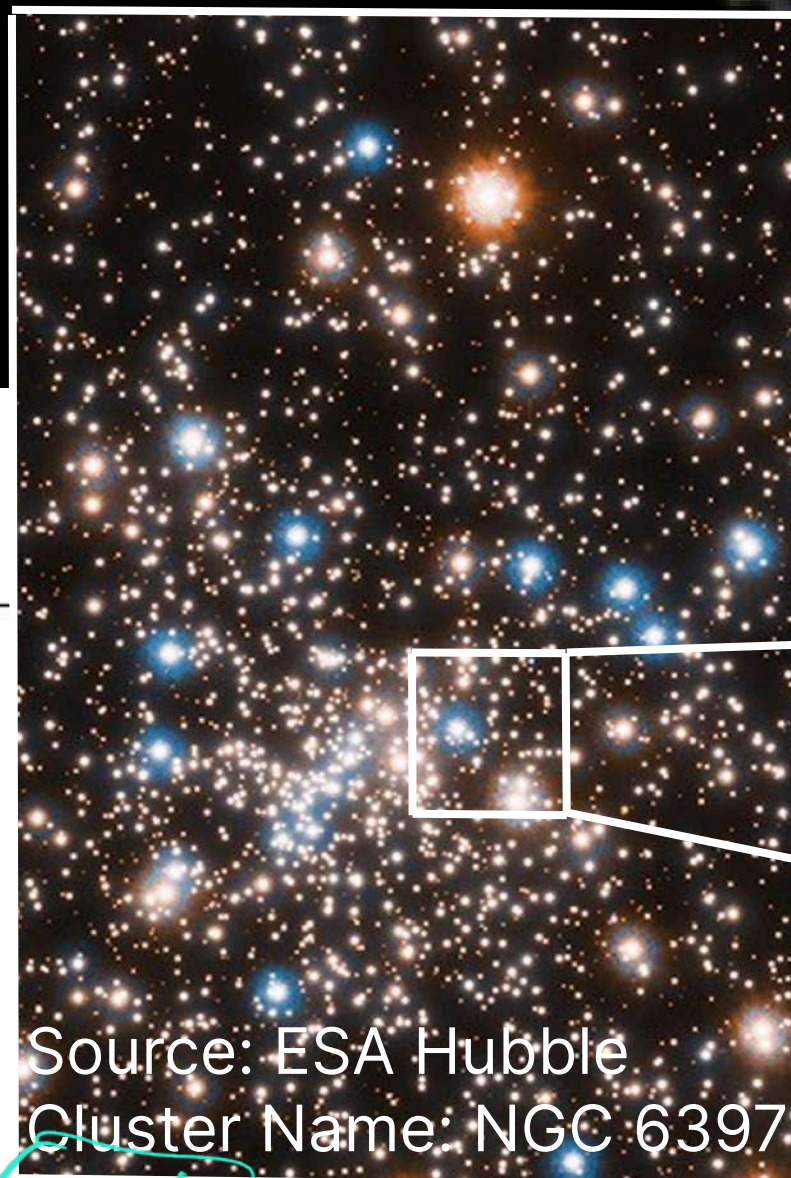
~~Let's dive into the Milky Way:~~

|                 |    |
|-----------------|----|
| Booklet Summary | 2  |
| Massive Stars   | 4  |
| Hot Stars       | 6  |
| Sun-Like Stars  | 8  |
| Red Dwarfs      | 10 |
| Red Giants      | 12 |
| White Dwarfs    | 14 |
| Protostars      | 16 |
| Supernovae      | 18 |
| Neutron Stars   | 20 |
| Black Holes     | 22 |
| The Milky Way   | 24 |
| Galaxies        | 26 |
| Bibliography    | 28 |

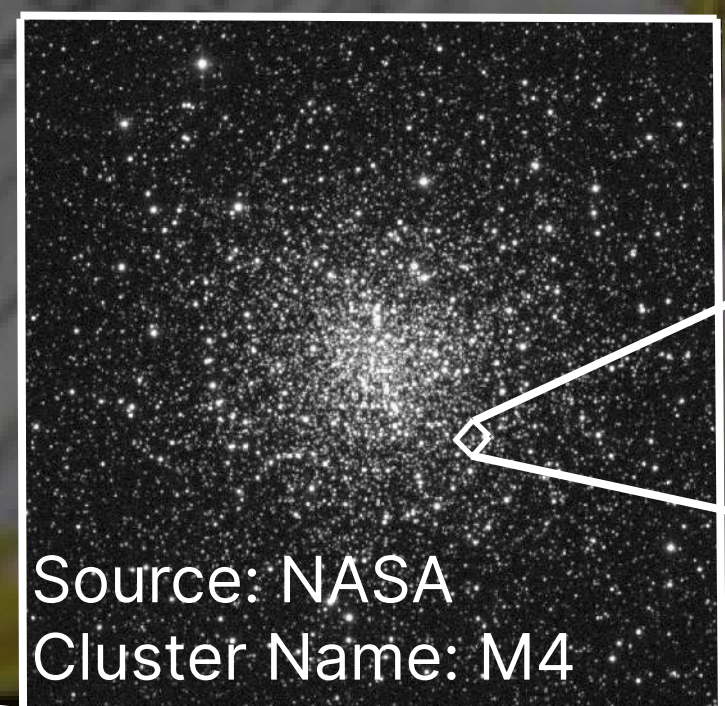
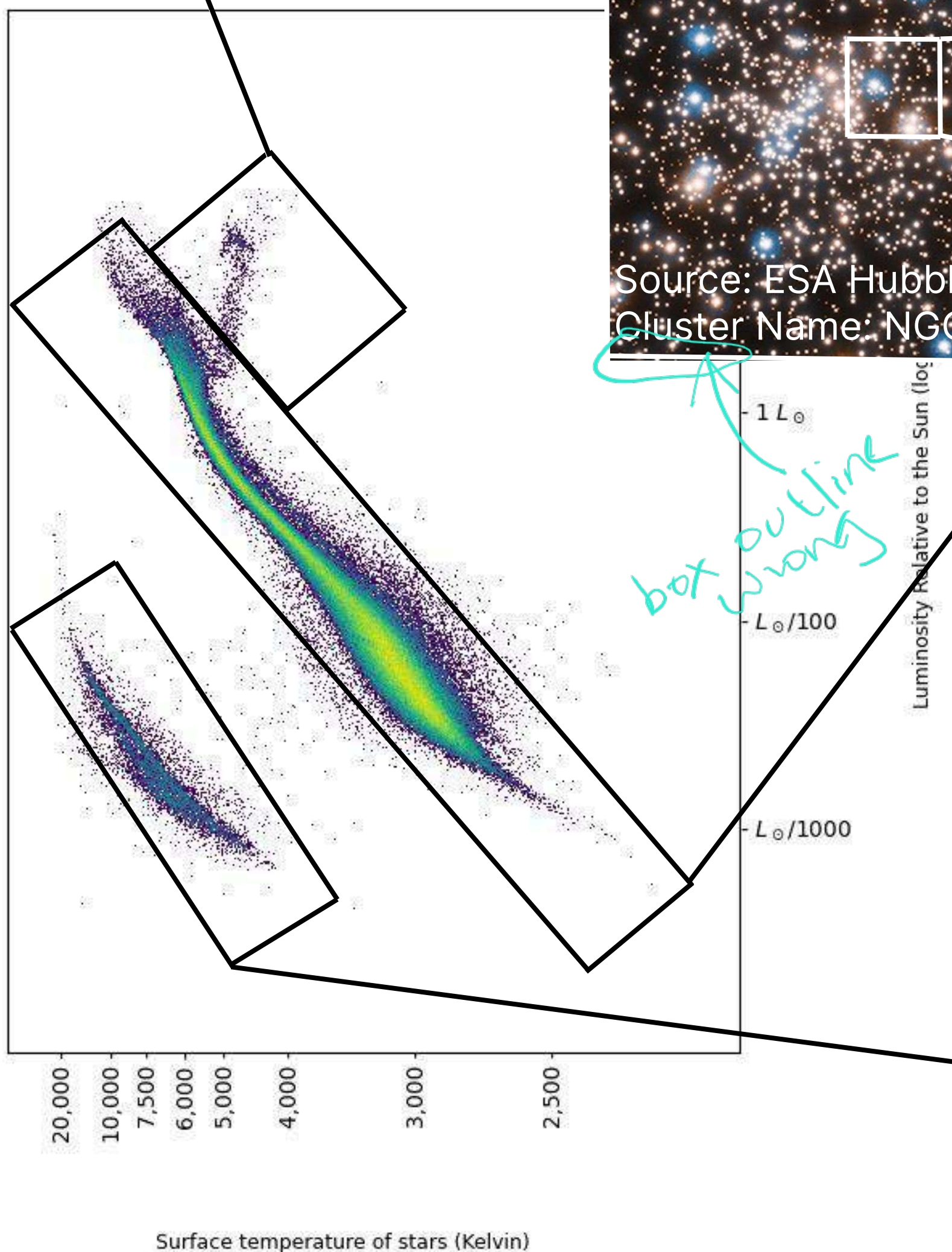




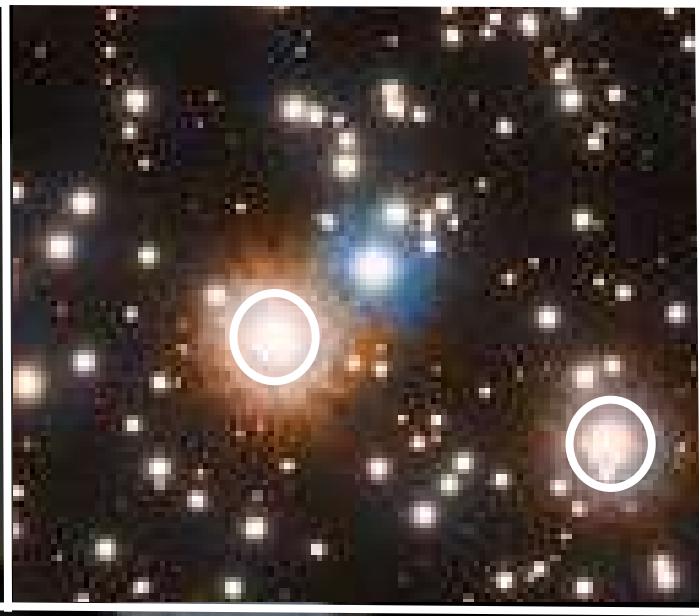
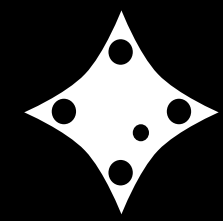
RED GIANTS



MAIN SEQUENCE



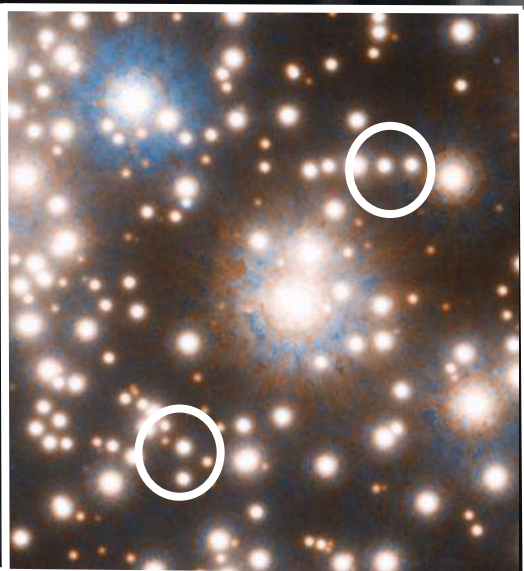




Humans have been admiring stars for thousands of years, yet there is so much more to learn about them beyond the brightnesses and colours we see.

A temperature-luminosity diagram (or Colour-Magnitude diagram) sorts stars into three regions: Giants, Main Sequence, and White Dwarfs (lower left).

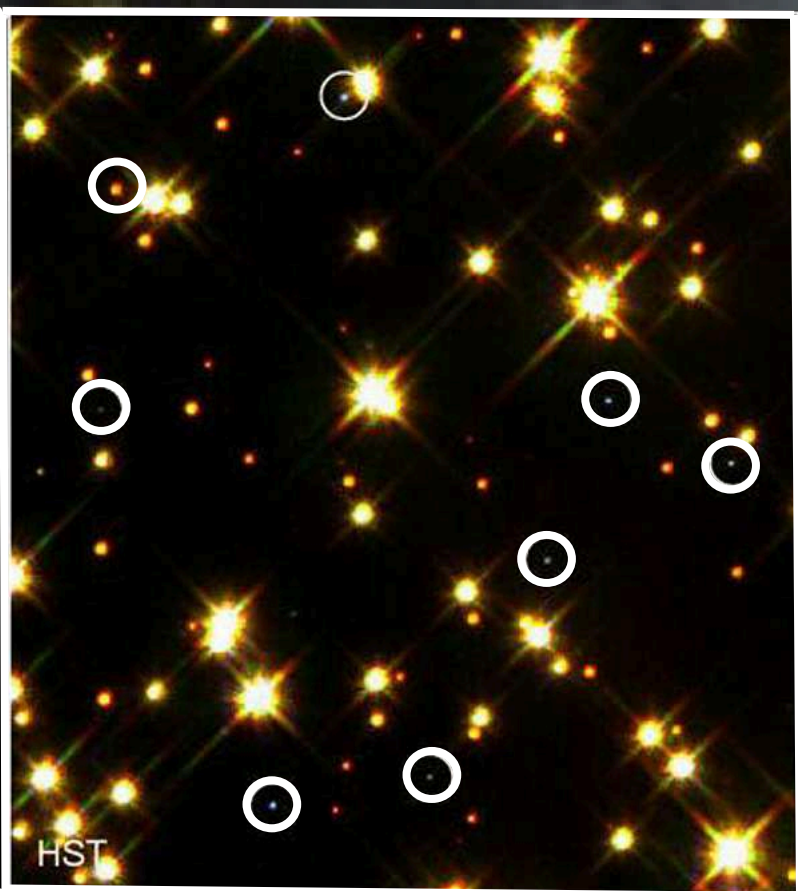
The GAIA telescope [background] measures the stars' colour, luminosity, and distance, plotting millions of new stars more accurately than ever before.



The images show stars in Globular Clusters, which consist up to a few 100,000 stars bound together through gravity.

-> could be worded better

WHITE  
DWARFS



Background Image Source: Gaia

smaller text

This booklet summarises properties of different stars, their fates in the long-term, and how they form galaxies.

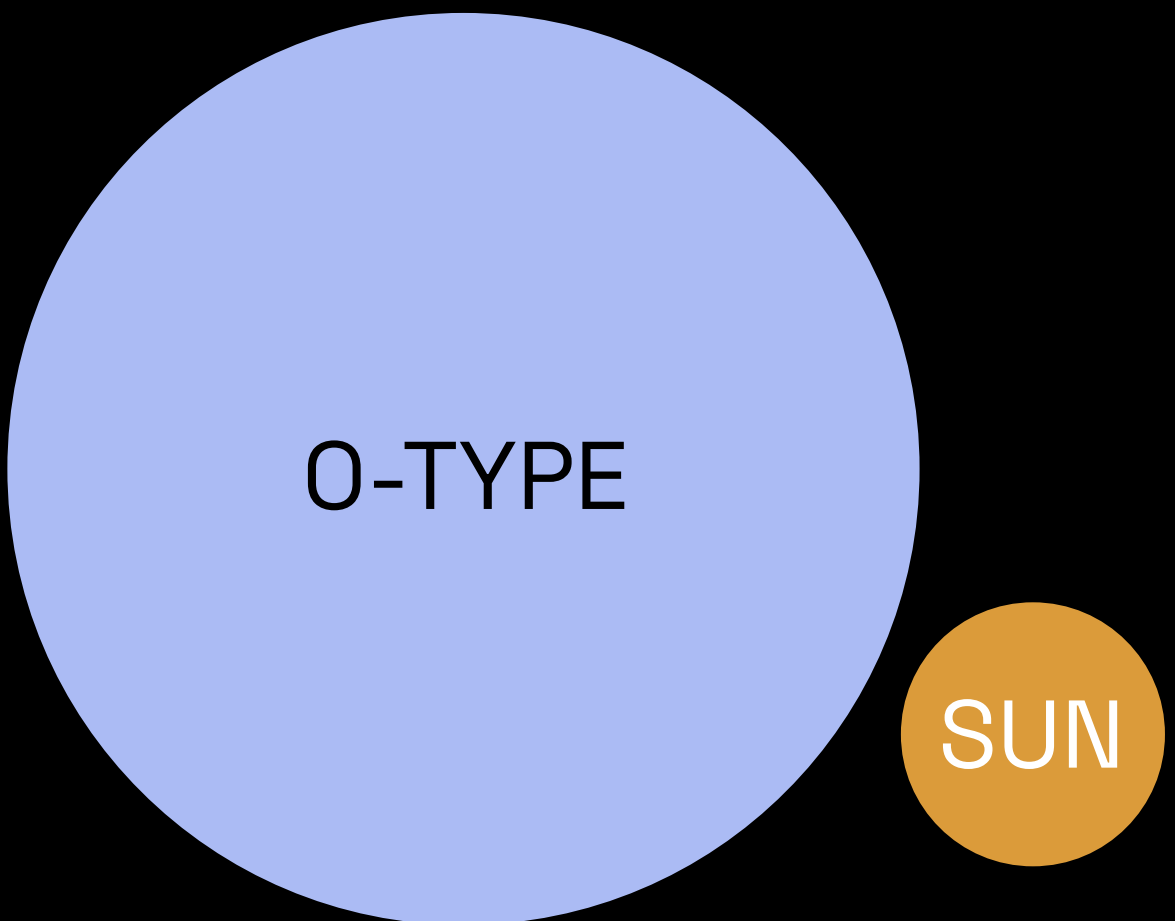




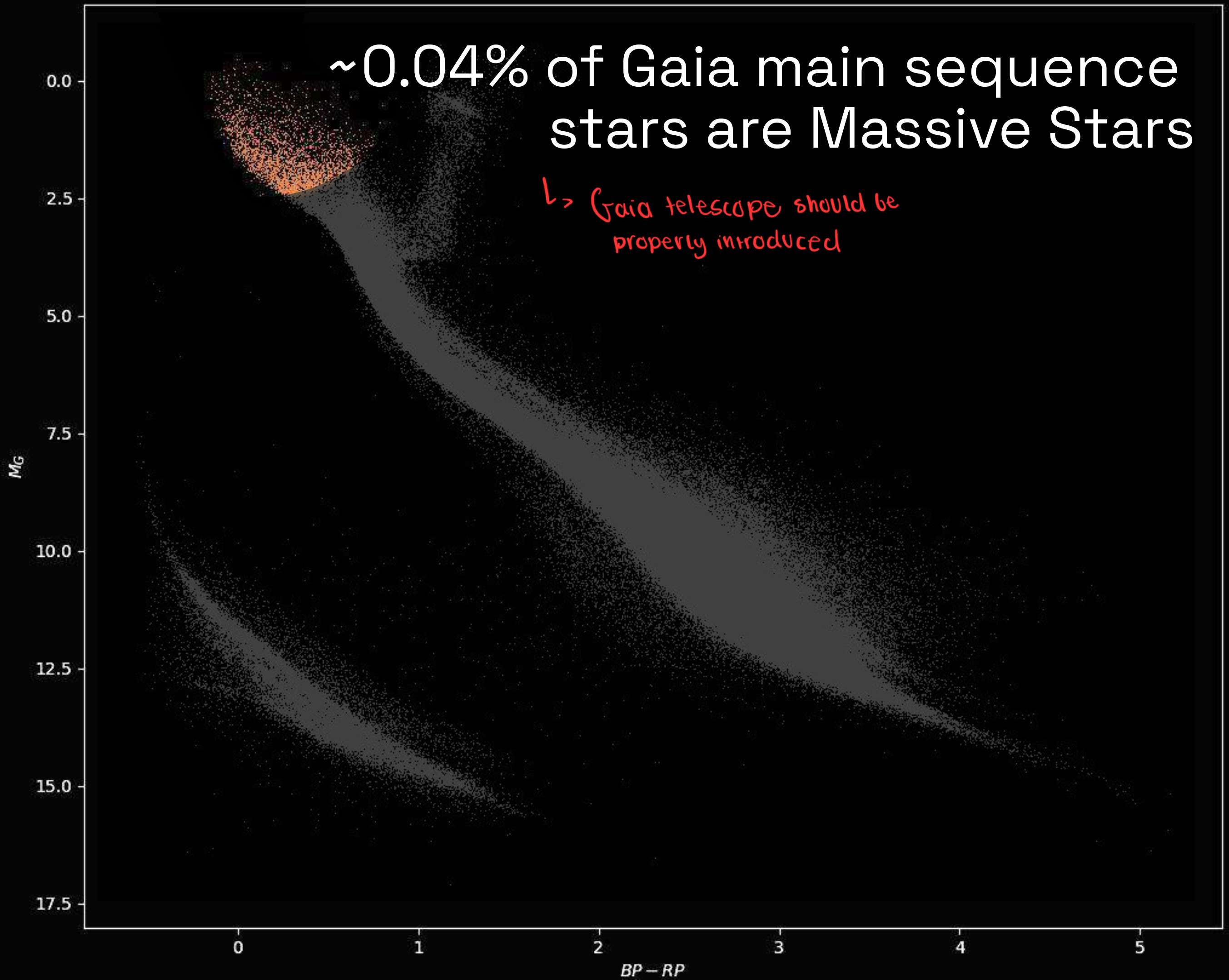
# MASSIVE STARS

Source: NASA

|                |                         |                                                                              |
|----------------|-------------------------|------------------------------------------------------------------------------|
| TYPE           | O and B                 | O- and B-type stars are the largest types of stars. They are blue in colour. |
| MASS           | 2-200 $M_{\odot}$       |                                                                              |
| TEMPERATURE    | 10000-50000 K           |                                                                              |
| LIFESPAN       | 10-100 million years    |                                                                              |
| RADIUS         | 2-10 $R_{\odot}$        |                                                                              |
| NEARBY EXAMPLE | $\zeta$ Ophiuchi        |                                                                              |
| LUMINOSITY     | 1000-100000 $L_{\odot}$ |                                                                              |



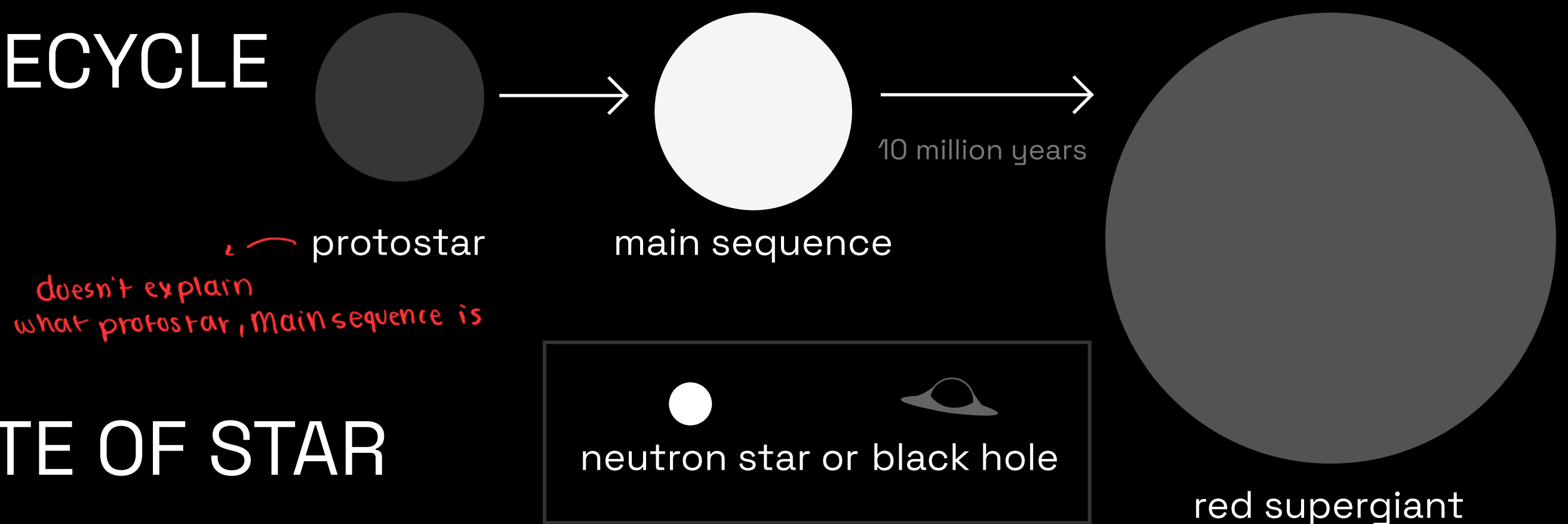




x-axis: colour

y-axis: brightness

## LIFECYCLE



## FATE OF STAR

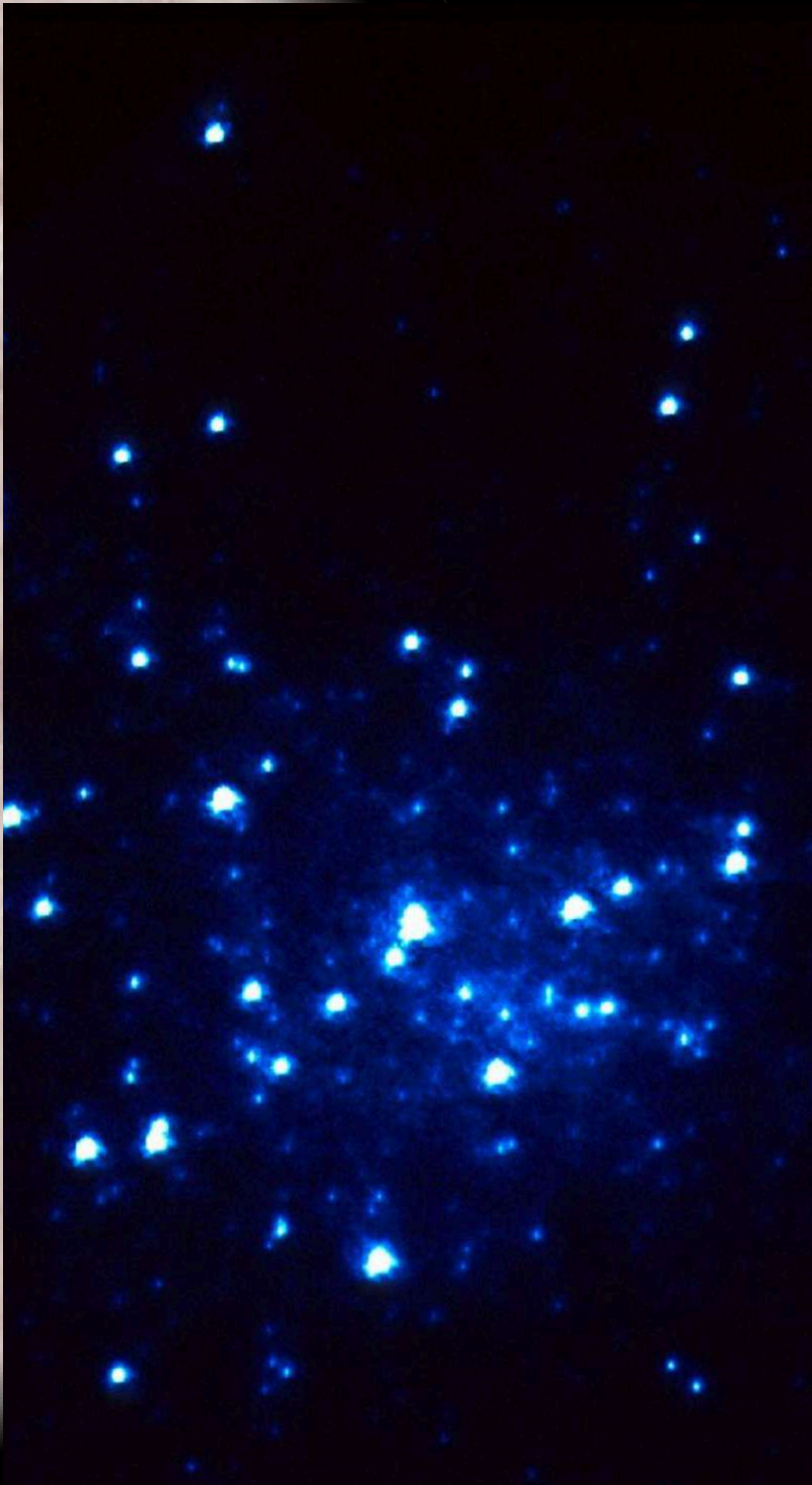
## ELEMENTS CREATED

|                                |                                |                                  |                                  |
|--------------------------------|--------------------------------|----------------------------------|----------------------------------|
| 79 196.97<br><b>Au</b><br>Gold | 82 207.20<br><b>Pb</b><br>Lead | 92 238.03<br><b>U</b><br>Uranium | 14 28.09<br><b>Si</b><br>Silicon |
|--------------------------------|--------------------------------|----------------------------------|----------------------------------|

*dull colour  
- inconsistent colour  
theme*



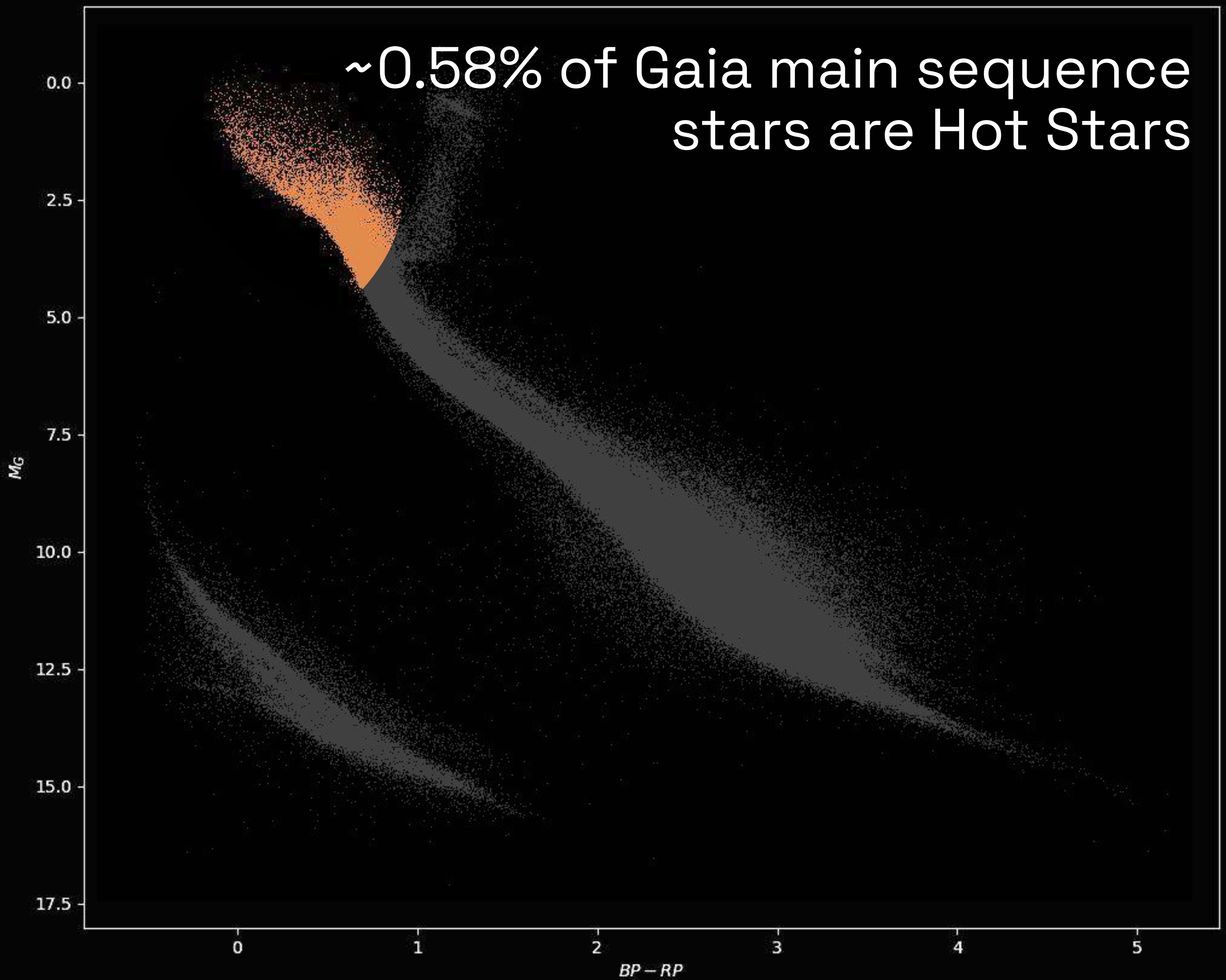
# HOT STARS



Source: Hubble Space Telescope

|             |                      |                                                                                                                                                                     |
|-------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TYPE        | A and F              | <p>A- and F-type stars are brighter and larger than our Sun. Their surface is hotter which is why they appear white.</p> <div><div>A-TYPE</div><div>SUN</div></div> |
| MASS        | 1.1-2.1 $M_{\odot}$  |                                                                                                                                                                     |
| TEMPERATURE | 6000-10000 K         |                                                                                                                                                                     |
| LIFESPAN    | 1-8 billion years    |                                                                                                                                                                     |
| RADIUS      | 1.15-1.8 $R_{\odot}$ |                                                                                                                                                                     |
| EXAMPLE     | Polaris              |                                                                                                                                                                     |
| LUMINOSITY  | 1.5-25 $L_{\odot}$   |                                                                                                                                                                     |

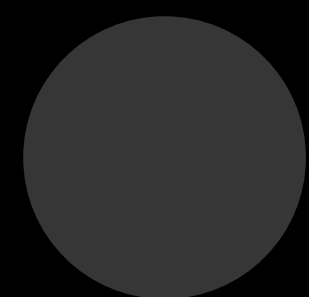




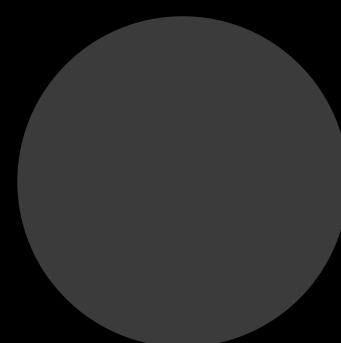
x-axis: colour

y-axis: brightness

## LIFECYCLE



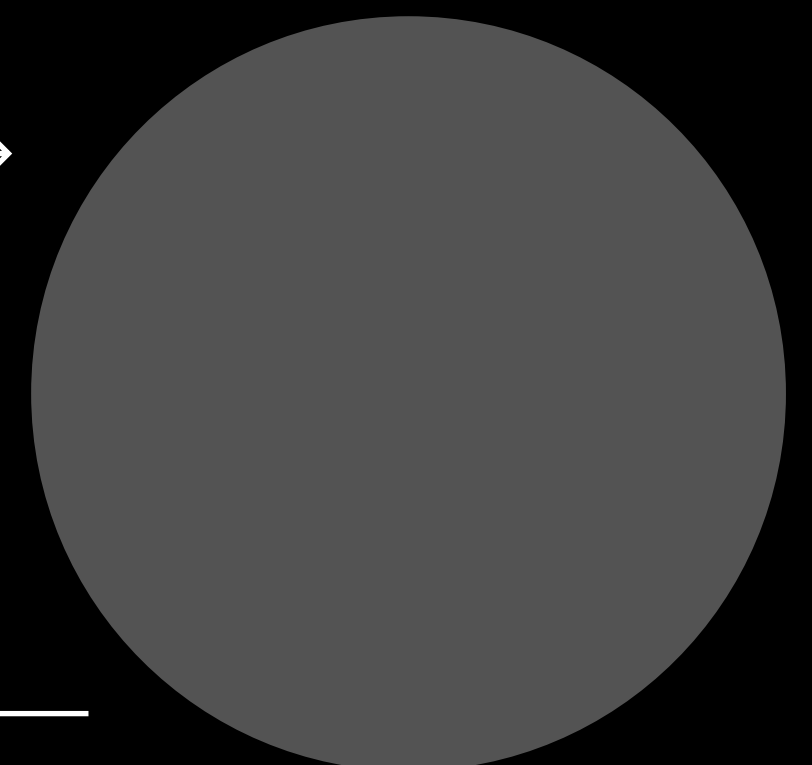
protostar



main sequence



10 billion years



red giant



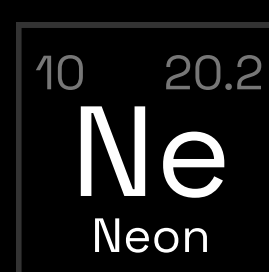
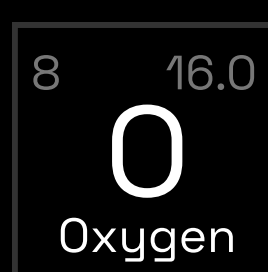
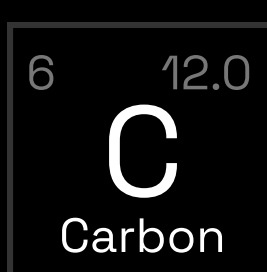
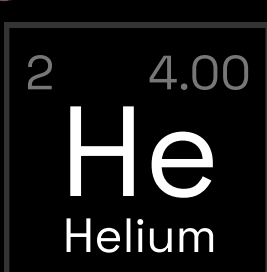
1 billion years



white dwarf

## FATE OF STAR

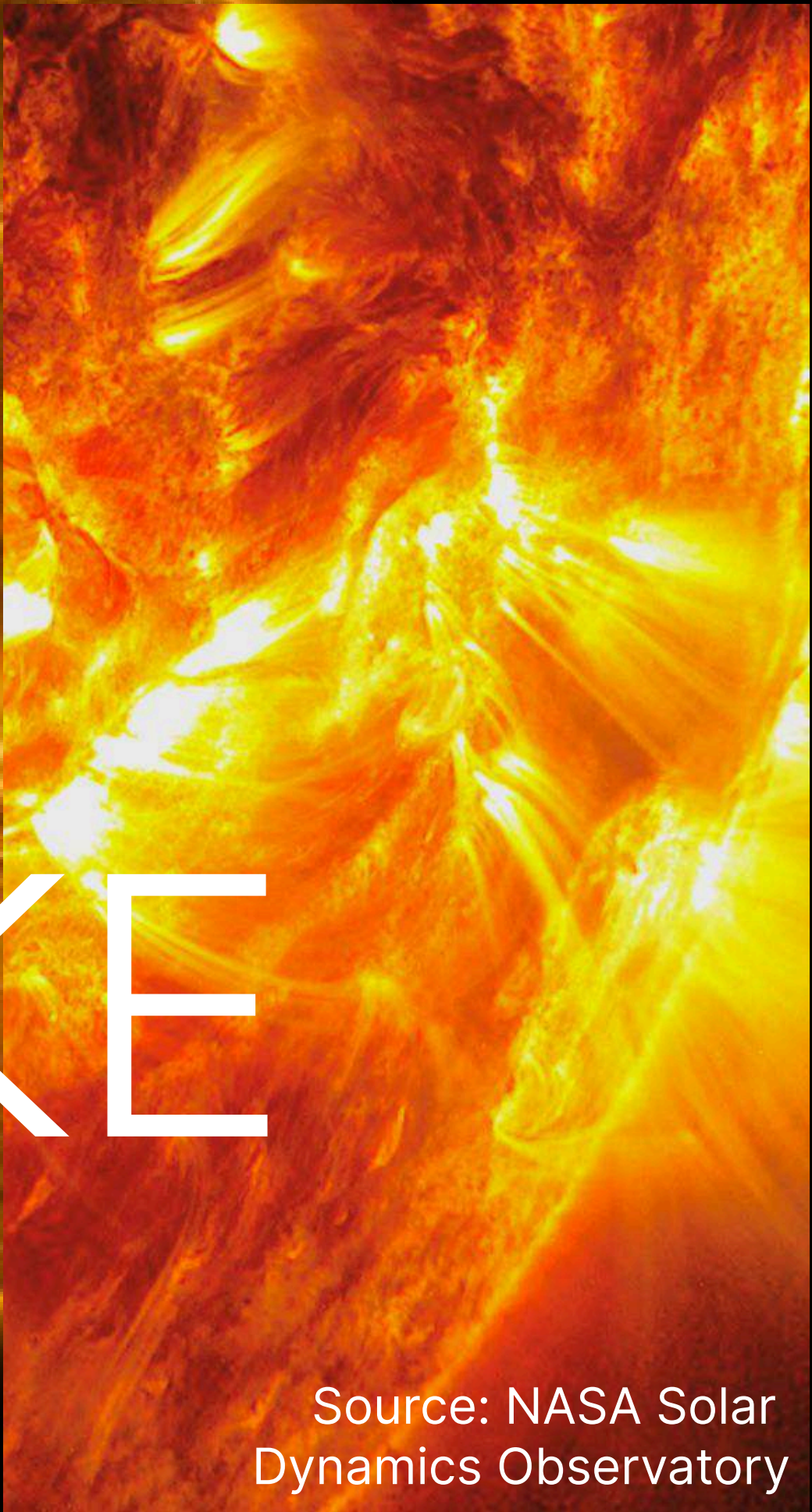
## ELEMENTS CREATED



Some A-type stars can explode in a supernova.

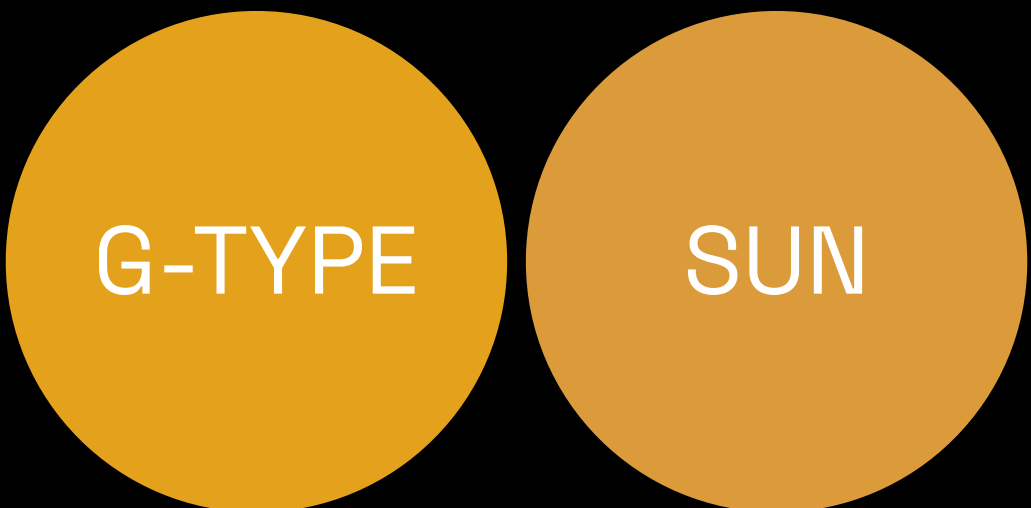


# SUN-LIKE STARS

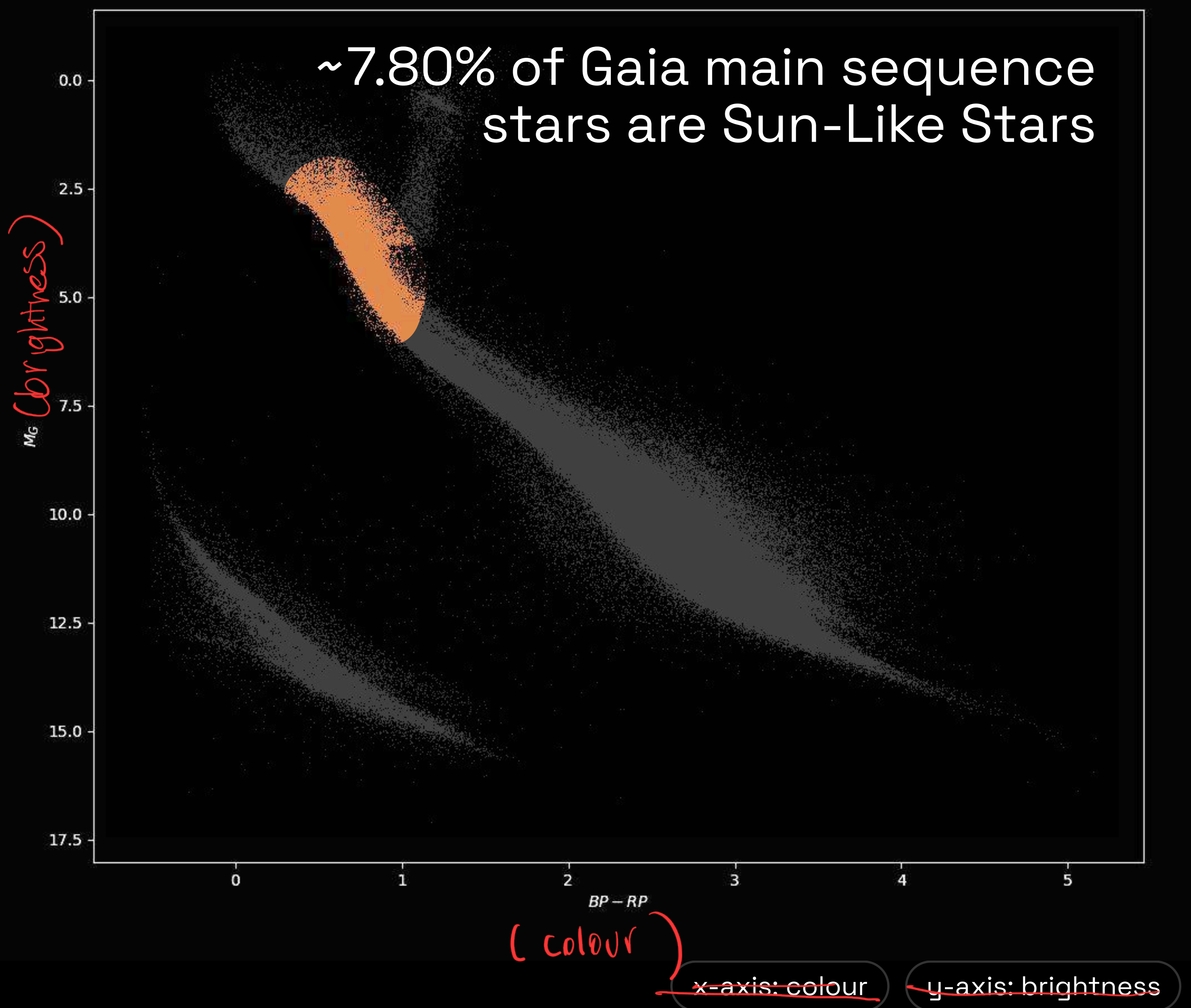
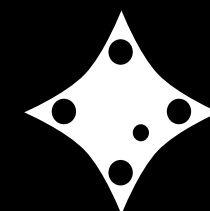


Source: NASA Solar Dynamics Observatory

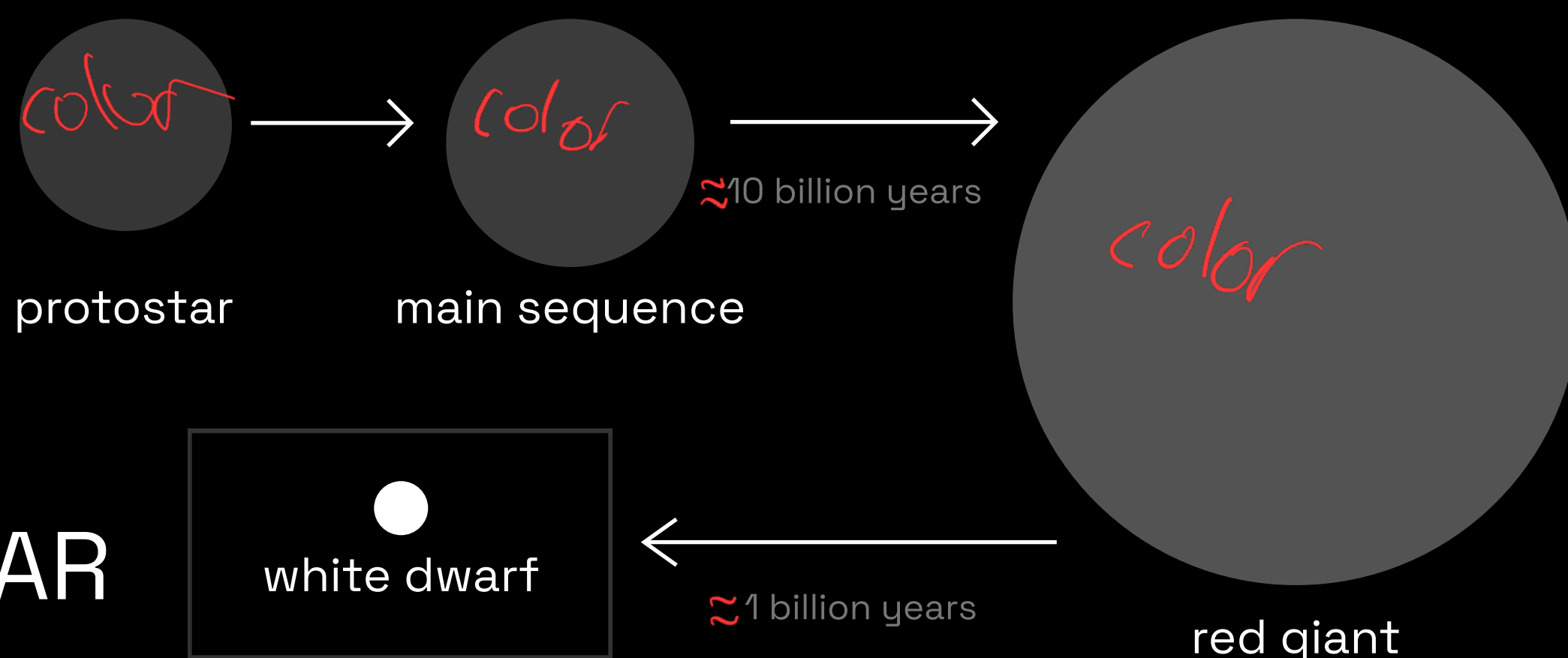
|                |                        |                                                                                                               |
|----------------|------------------------|---------------------------------------------------------------------------------------------------------------|
| TYPE           | G                      | <p>G-type stars are relatively stable yellow stars. Our Sun is categorised as a G-type star!</p> <p>align</p> |
| MASS           | 0.9-1.1 $M_{\odot}$    |                                                                                                               |
| TEMPERATURE    | 5000 - 6000 K          |                                                                                                               |
| LIFESPAN       | 7.9 - 20 billion years |                                                                                                               |
| RADIUS         | 0.8 - 1.1 $R_{\odot}$  |                                                                                                               |
| NEARBY EXAMPLE | The Sun                |                                                                                                               |
| LUMINOSITY     | 0.6 - 1.5 $L_{\odot}$  |                                                                                                               |







## LIFECYCLE



## FATE OF STAR

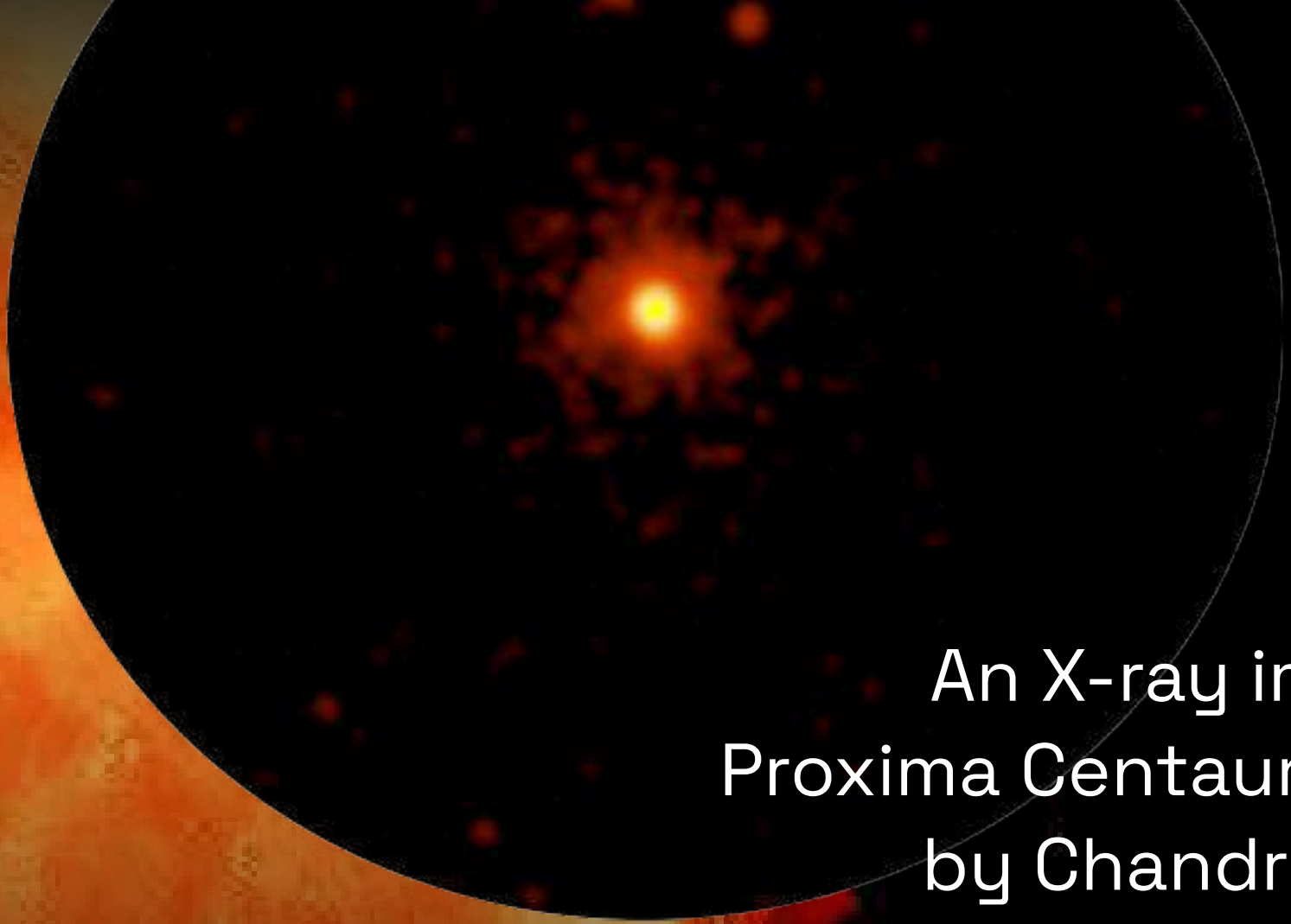
## ELEMENTS CREATED

|        |      |        |      |        |      |
|--------|------|--------|------|--------|------|
| 2      | 4.00 | 6      | 12.0 | 8      | 16.0 |
| He     |      | C      |      | O      |      |
| Helium |      | Carbon |      | Oxygen |      |

The Sun's corona is  $\sim 200\times$  hotter than its surface - no-one knows why!

create divide



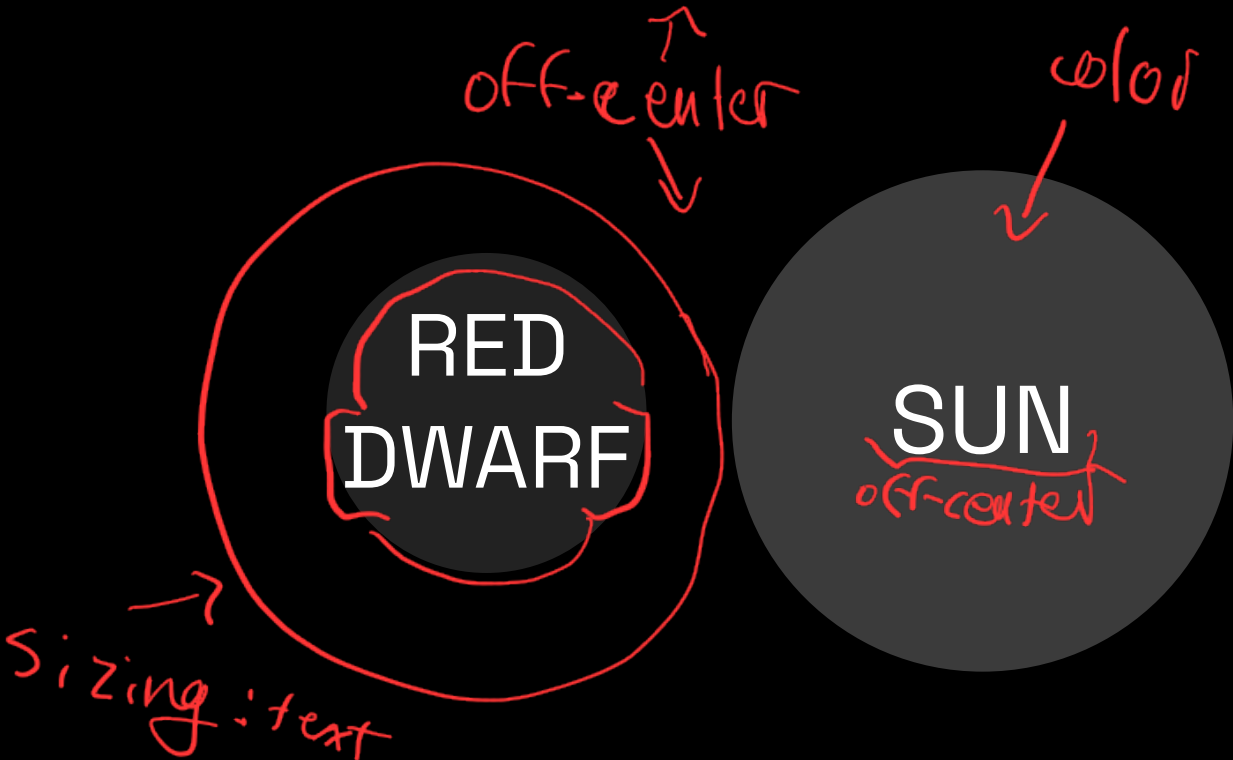


An X-ray image of Proxima Centauri, taken by Chandra X-Ray Observatory

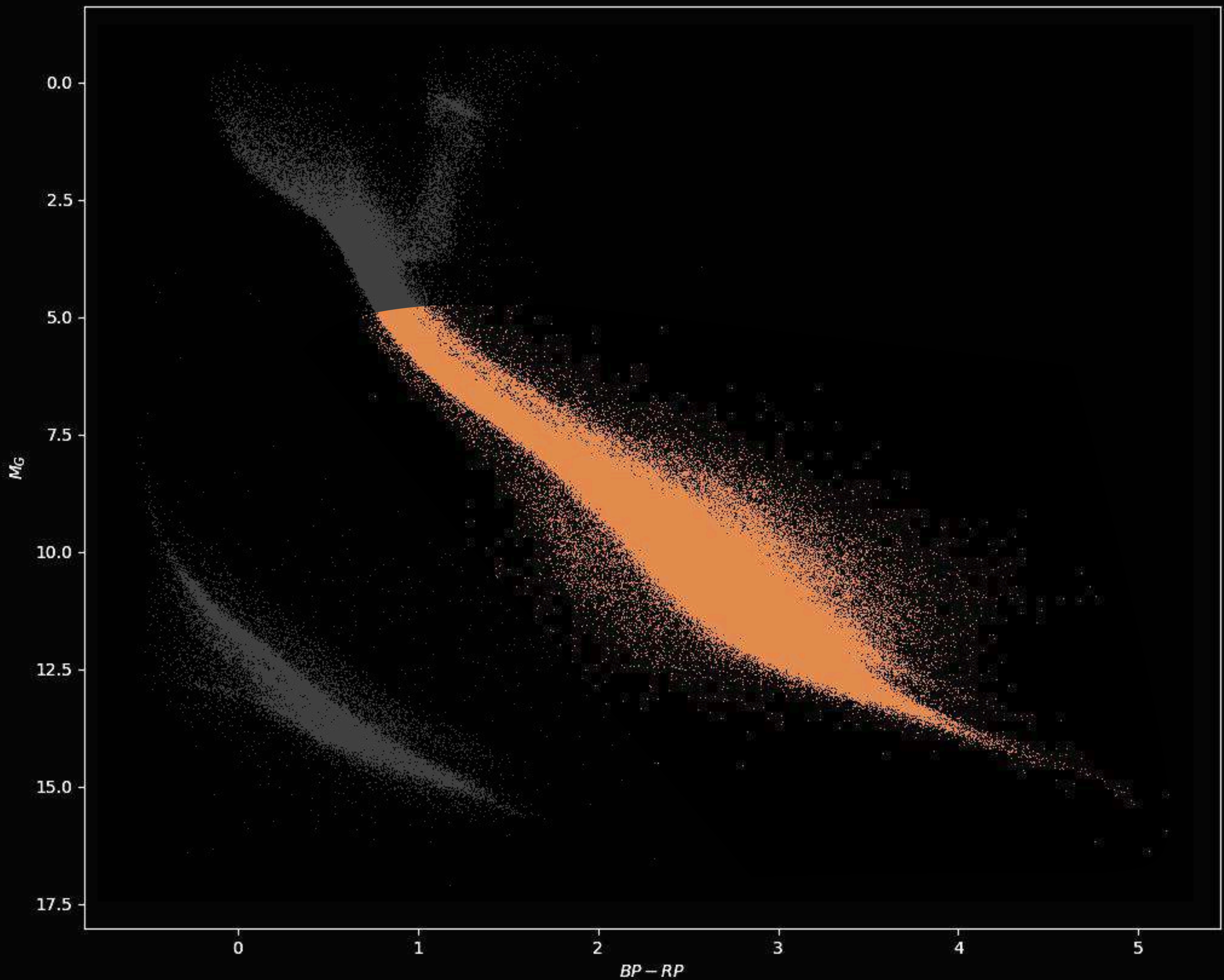
# RED DWARFS

|             |                          |
|-------------|--------------------------|
| TYPE        | K and M                  |
| MASS        | 0.08 - 0.9 $M_{\odot}$   |
| TEMPERATURE | 2500 - 5300 K            |
| LIFESPAN    | 100+ billion years       |
| RADIUS      | 0.06 - 1.08 $R_{\odot}$  |
| EXAMPLE     | Proxima Centauri         |
| LUMINOSITY  | 0.0001 - 0.5 $L_{\odot}$ |

M type stars are the most common in our universe. They live upwards of 100 billion years and burn a red-orange colour.



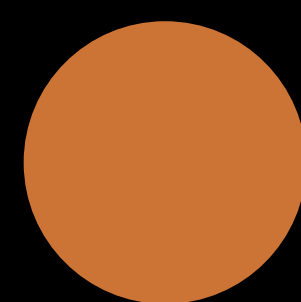




x-axis: colour

y-axis: brightness

## LIFECYCLE



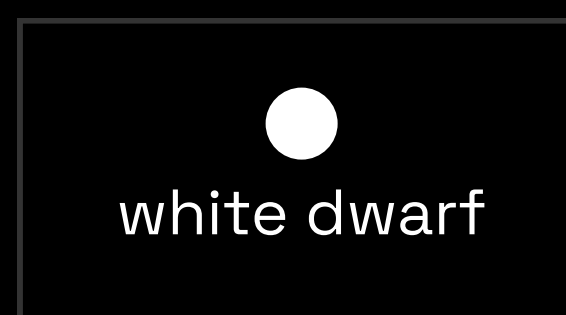
protostar



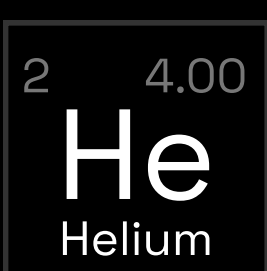
main sequence

100 billion years

## FATE OF STAR



## ELEMENTS CREATED



Even as the most common star type, making up 70-80% of the Milky Way, they are invisible to the naked eye





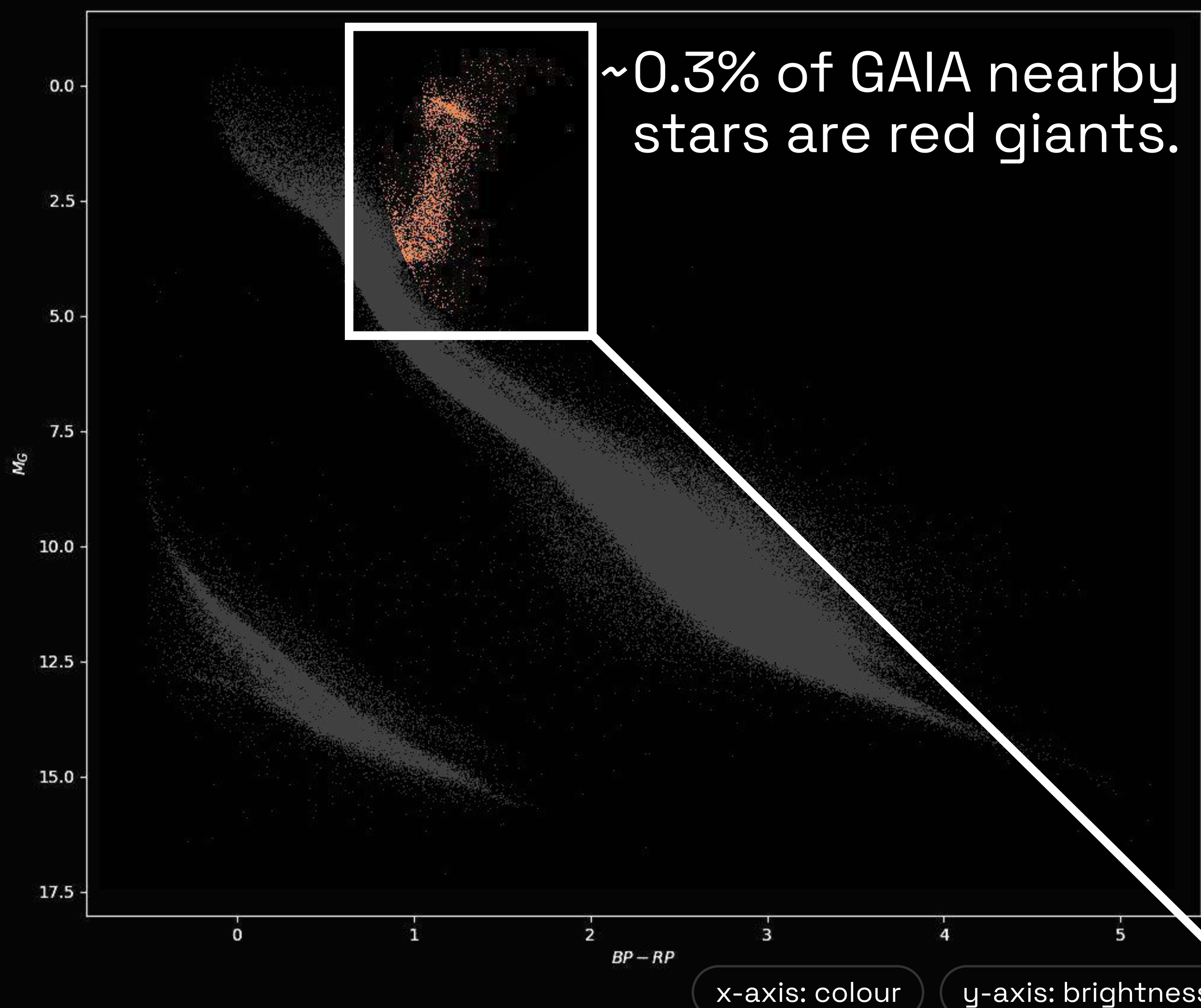
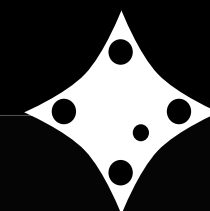
# RED GIANTS



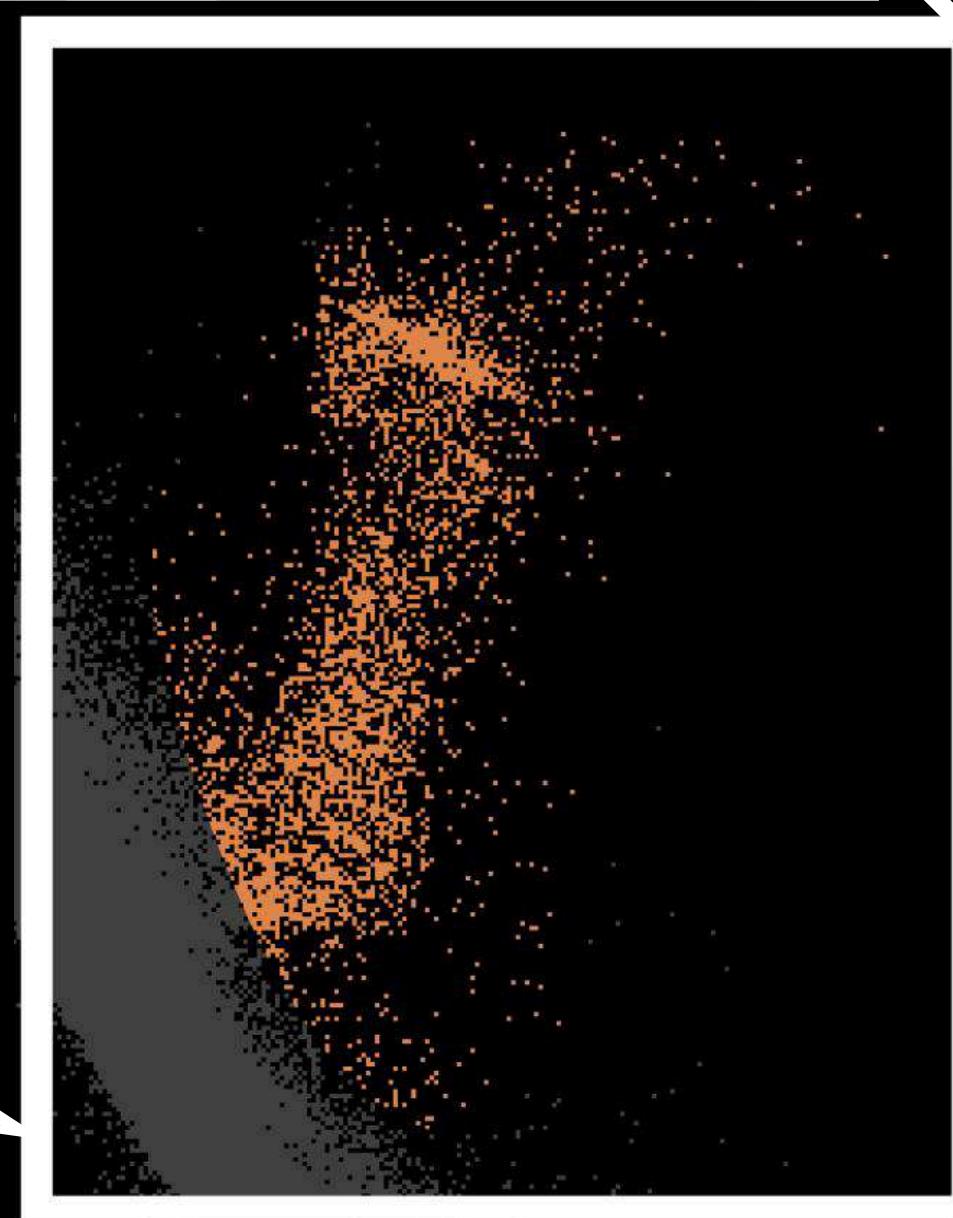
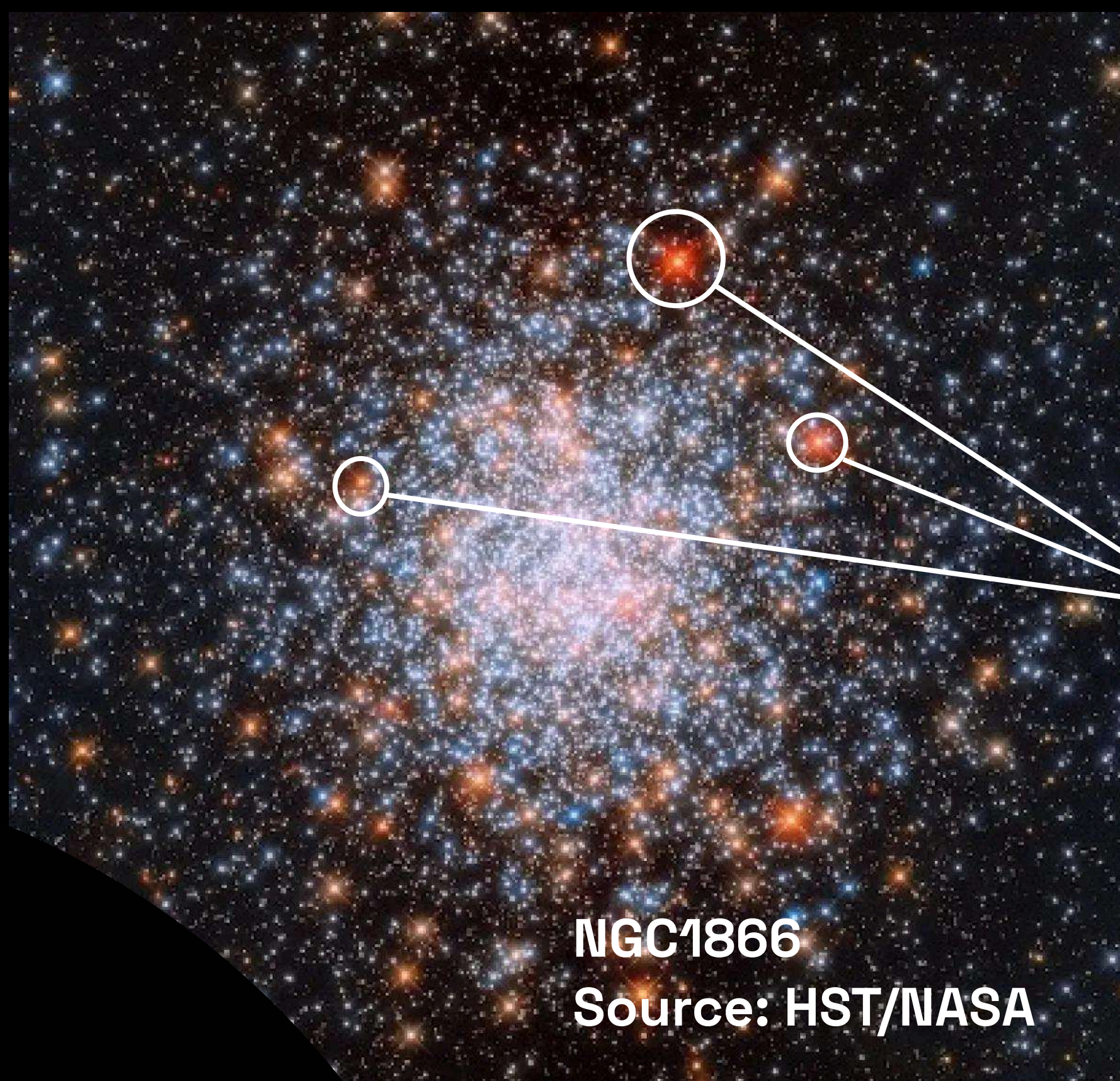
Source: : European Southern Observatory

|             |                           |                                                              |                                                           |                                                           |
|-------------|---------------------------|--------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|
| TYPE        | RGB, HB and AGB           | ELEMENTS FOUND                                               |                                                           |                                                           |
| MASS        | $0.5-5 M_{\odot}$         | <div><div>1428.09</div><div>Si</div><div>Silicon</div></div> | <div><div>612.00</div><div>C</div><div>Carbon</div></div> | <div><div>5655.85</div><div>Fe</div><div>Iron</div></div> |
| TEMPERATURE | 5000-3000 K               |                                                              |                                                           |                                                           |
| LIFESPAN    | $10^7$ to $10^{10}$ years |                                                              |                                                           |                                                           |
| RADIUS      | $\sim 100 R_{\odot}$      | SUN                                                          |                                                           |                                                           |
| EXAMPLE     | Antares                   |                                                              |                                                           |                                                           |
| LUMINOSITY  | $\sim 3000 L_{\odot}$     | RED GIANT                                                    |                                                           |                                                           |





## Bright Red Giants in a Globular Cluster



Red in colour caused by cooler surfaces, and temperatures than the Sun



# WHITE DWARFS

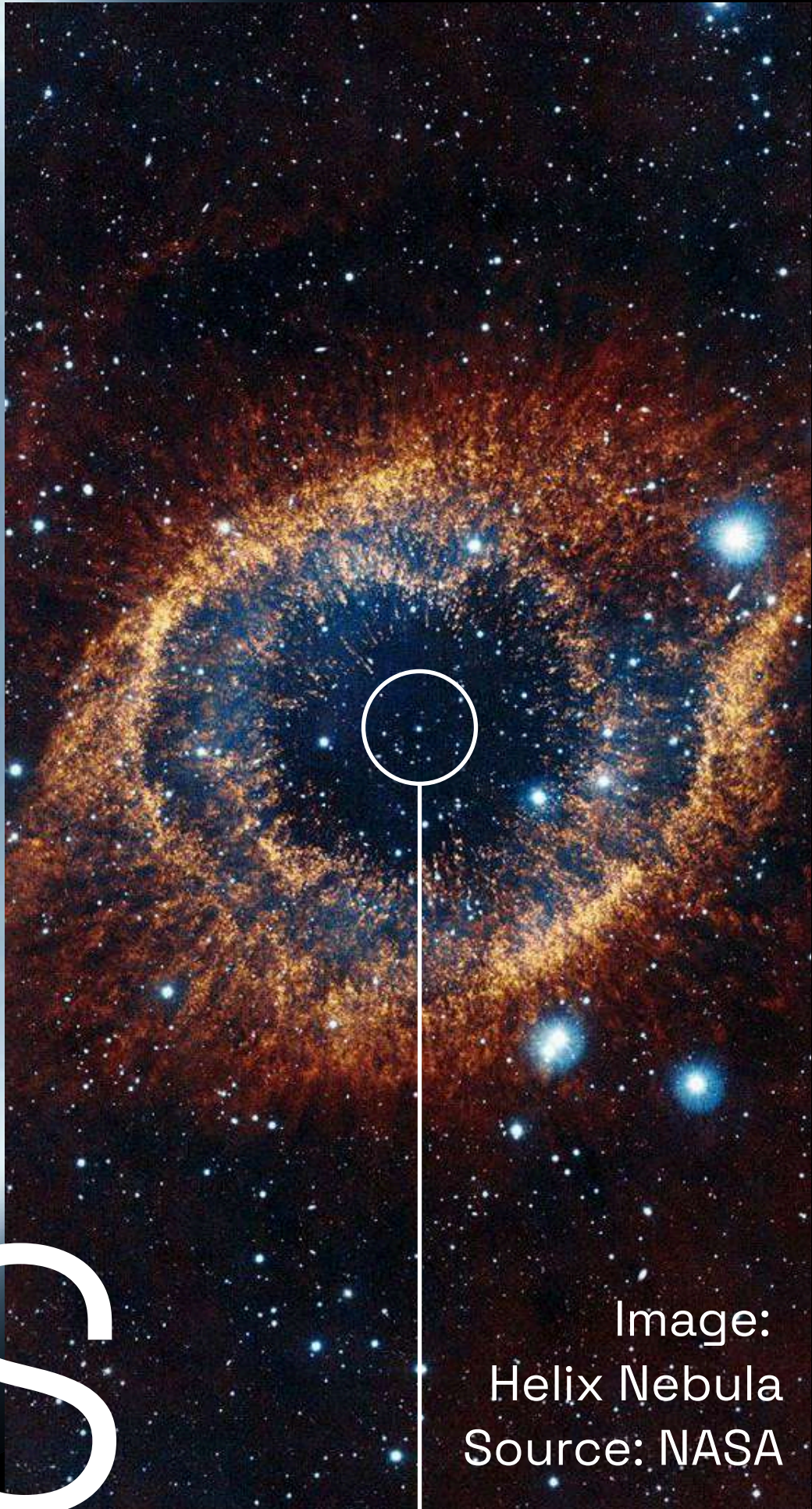


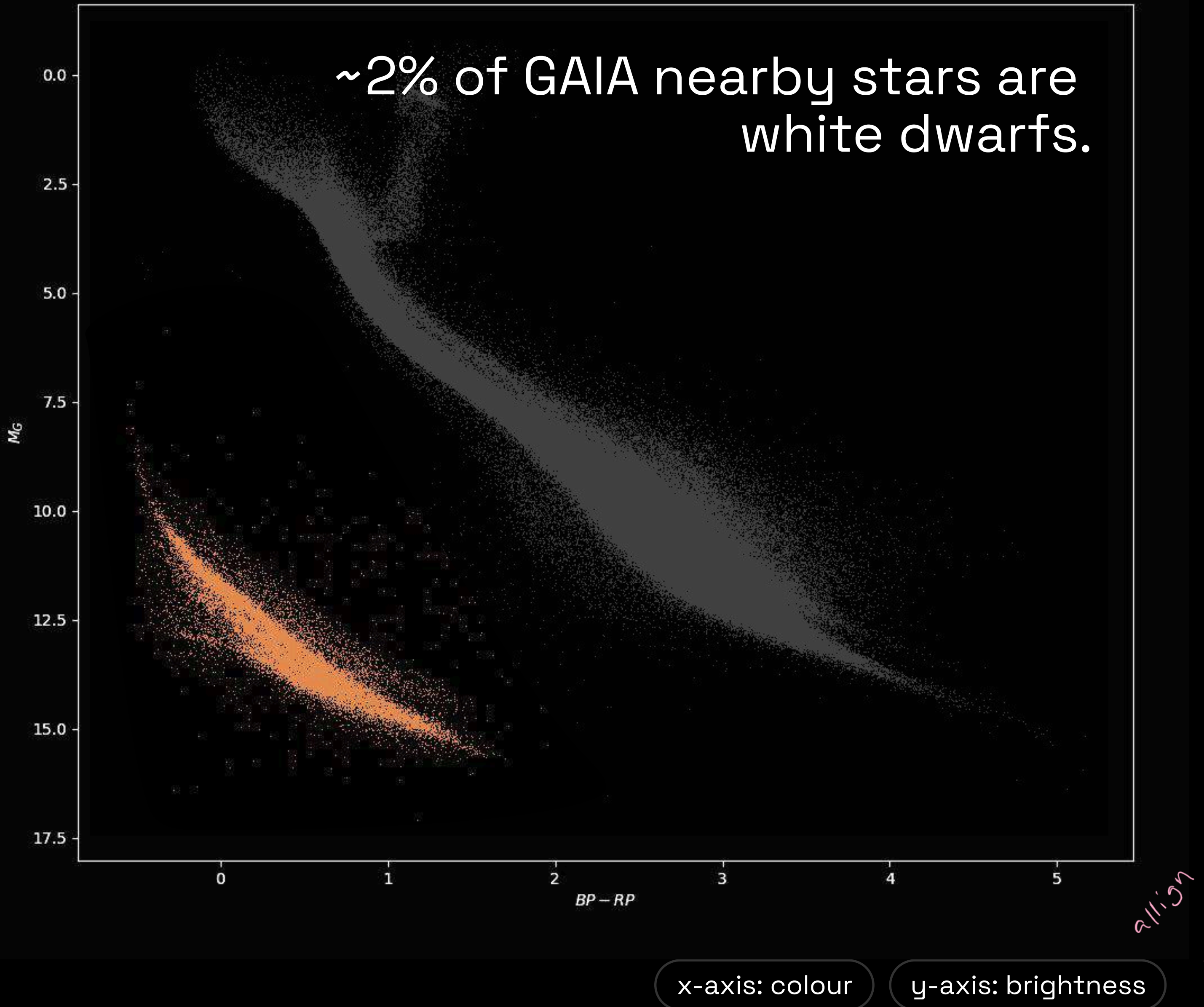
Image:  
Helix Nebula  
Source: NASA

|                |                     |
|----------------|---------------------|
| TYPE           | He, CO and ONeMg    |
| MASS           | 0.2-1.3 $M_{\odot}$ |
| TEMPERATURE    | 8000 - 40000 K      |
| LIFESPAN       | 100+ billion years  |
| RADIUS         | 6000-7000km         |
| NEARBY EXAMPLE | Sirius B            |
| LUMINOSITY     | 0.1 - 1 $L_{\odot}$ |

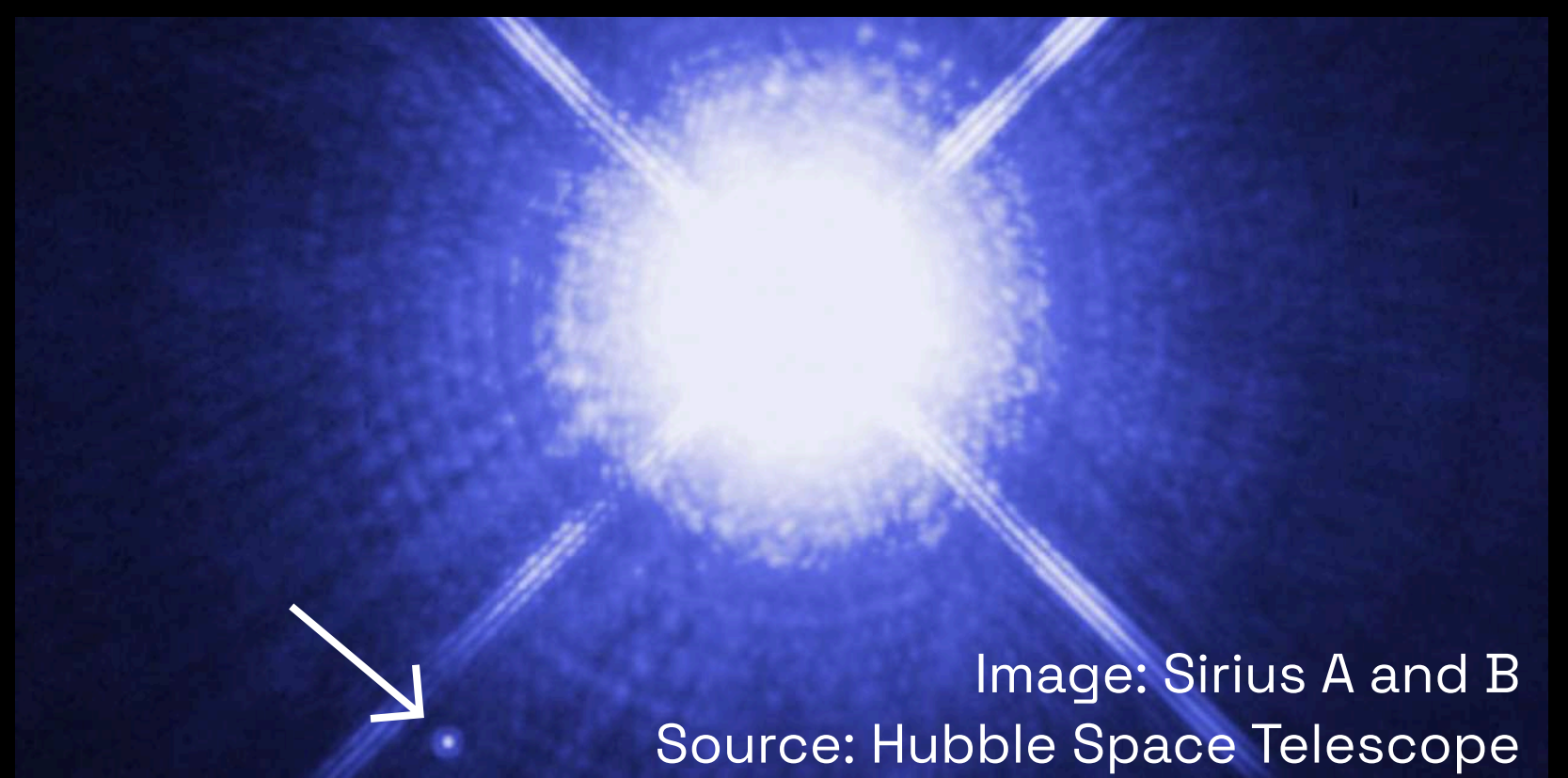
White dwarfs are the leftover dead cores of intermediate mass stars like our Sun. An example is the faint star in the centre of the Helix Nebula, shown above.







White dwarfs are often denoted as “He”, “CO”, or “ONeMg”, based on their chemical composition after the red giant phase of stellar evolution.



Sirius A (highly luminous star) and Sirius B (tiny dot). Sirius B is a white dwarf. *needs to be smaller text*

## ELEMENTS FOUND

|           |      |          |      |          |      |           |      |
|-----------|------|----------|------|----------|------|-----------|------|
| 2         | 4.00 | 6        | 12.0 | 8        | 16.0 | 10        | 20.2 |
| <b>He</b> |      | <b>C</b> |      | <b>O</b> |      | <b>Ne</b> |      |
| Helium    |      | Carbon   |      | Oxygen   |      | Neon      |      |

One teaspoon of a white dwarf can weigh as much as **5 tons.**

*overlapping*





ASTRAL

Source: NASA/ESA/CSA/STScI

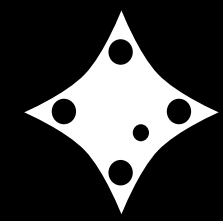
# PROTOSTARS

Protostars form  
in the center of  
**GAS CLOUDS**  
that are over **100x**  
the size of the star

Jets of remaining gas are  
funnelled outwards by  
**MAGNETIC  
WINDS**

The gas clouds can be  
seen along the spiral  
arms of the Milky Way





# PROTOSTARS

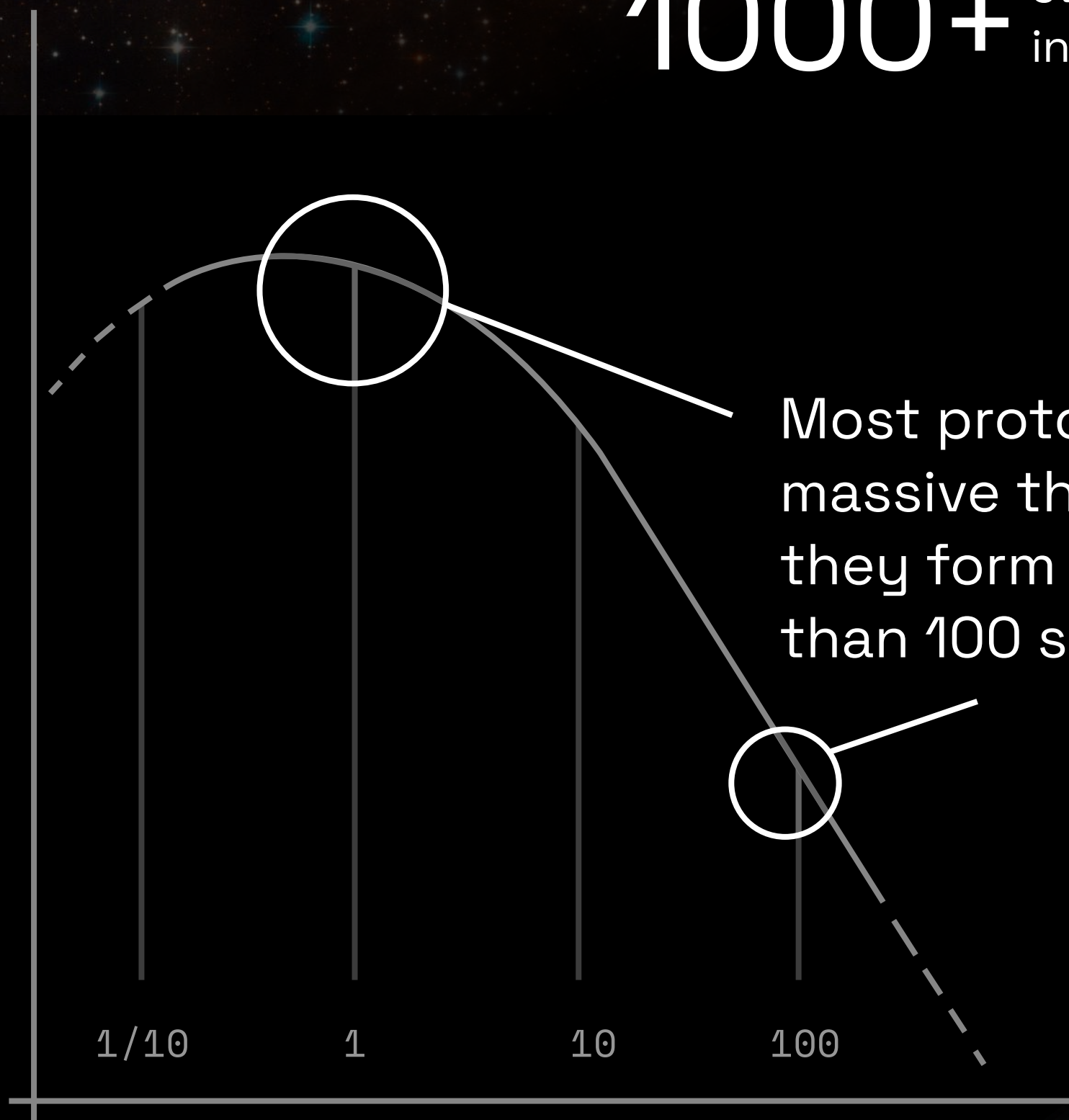


Protostars are stars that are just beginning to form. At this stage, they are not yet large enough to fuse hydrogen atoms into helium.

Stars usually form in  
**CLUSTERS**  
around large clouds of gas;

**1000+** stars can form  
in the same area

Number of protostars

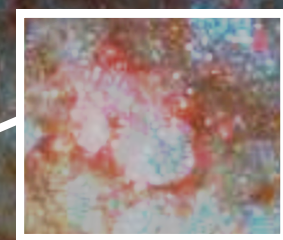


Most protostars are less massive than the Sun when they form - very few are heavier than 100 solar masses

Mass of protostar (solar masses)

Hydrogen is ionised by the radiation from new stars, which can be seen as  
**PINK LIGHT** in areas where protostars are forming.

Whirlpool galaxy  
Source: HST/NASA



*smaller*





Supernovae are **BRIGHT**, *hyper-energetic* explosions that occur with violent star deaths. They are the most powerful explosions ever discovered.

They are bright enough that they can outshine their **host galaxy**, and people have been recording supernova sightings since before the telescope was invented.

not important enough to bold

too many effects

highlight in some way what exactly is being talked about

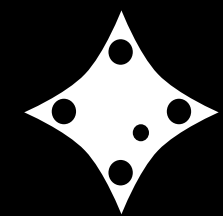
Supernovae are not very common, only occurring two or three times a century per galaxy, so modern scientists will only observe a few hundred supernovae from outside the Milky Way. Space dust obscures our view of supernovae inside our galaxy, making them relatively more difficult to see.

not aligned

These explosions are powerful enough to create extremely **heavy elements**, including those crucial for the formation of life. The massive, violent explosions distribute them across the universe, scattering both the existing contents of the star and the newly created elements.



There are two main types of supernovae: **Type I** and **Type II**



ASTRAL

# TYPE I

off centre

**Type I** supernovae occur in binary systems where one of the stars is a **WHITE DWARF**.

These are the **RAREST** type of supernovae and occur around once every **500 YEARS**.

Type I supernovae occur when a white dwarf **EXPLODES** because it sucks too much material away from its companion star.

uneventful

changes size too much

The G299 Type Ia supernova remnant  
Source: NASA/CXC/U.Texas

# TYPE II

**Type II** supernovae originate from the collapse and explosion of **MASSIVE STARS**.

As massive stars burn very hot, they quickly run out of fuel. The outward pressure generated at its core weakens without fuel, allowing gravity pushing inward to overpower it. The temperature drops, and the star collapses. Lasting only seconds, the implosion creates huge shock waves that rip the outer layers apart, leaving behind only a gaseous surrounding nebula and the dense core at its centre. In extremely massive stars, it may leave behind a black hole.

**The Crab Nebula Type II Supernova Remnant** first seen in 1054 AD  
**Source: NASA**

remove shadows

centre text

MAJOR  
SUPERNOVAE





ASTRAL

Neutron stars are about the size of Melbourne, but weigh as much as

1.4 SUNS!

"The Nebula around the Crab Nebula"

A spoonful of neutron stars weighs as much as a

MOUNTAIN

When a star

COLLAPSES

on itself during a supernova, its protons and electrons are squeezed together into a

SUPER DENSE

NEUTRON STAR

Source:  
NASA/CXC/A. Hobart

remove or  
move to right

Some spinning neutron stars point towards Earth, or have an extra strong

MAGNETIC FIELD

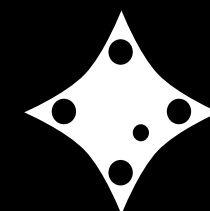
We call these

PULSARS and MAGNETARS respectively.

Source:  
Lizel Arina, Freepik

addicted  
✓ or caps





## NEUTRON STARS

A pulsar is a rapidly-rotating neutron star that has a lighthouse-like beam of emission. When the beam crosses the Earth, we detect regular pulses of emission, hence the name “pulsar”.



Fun Fact:

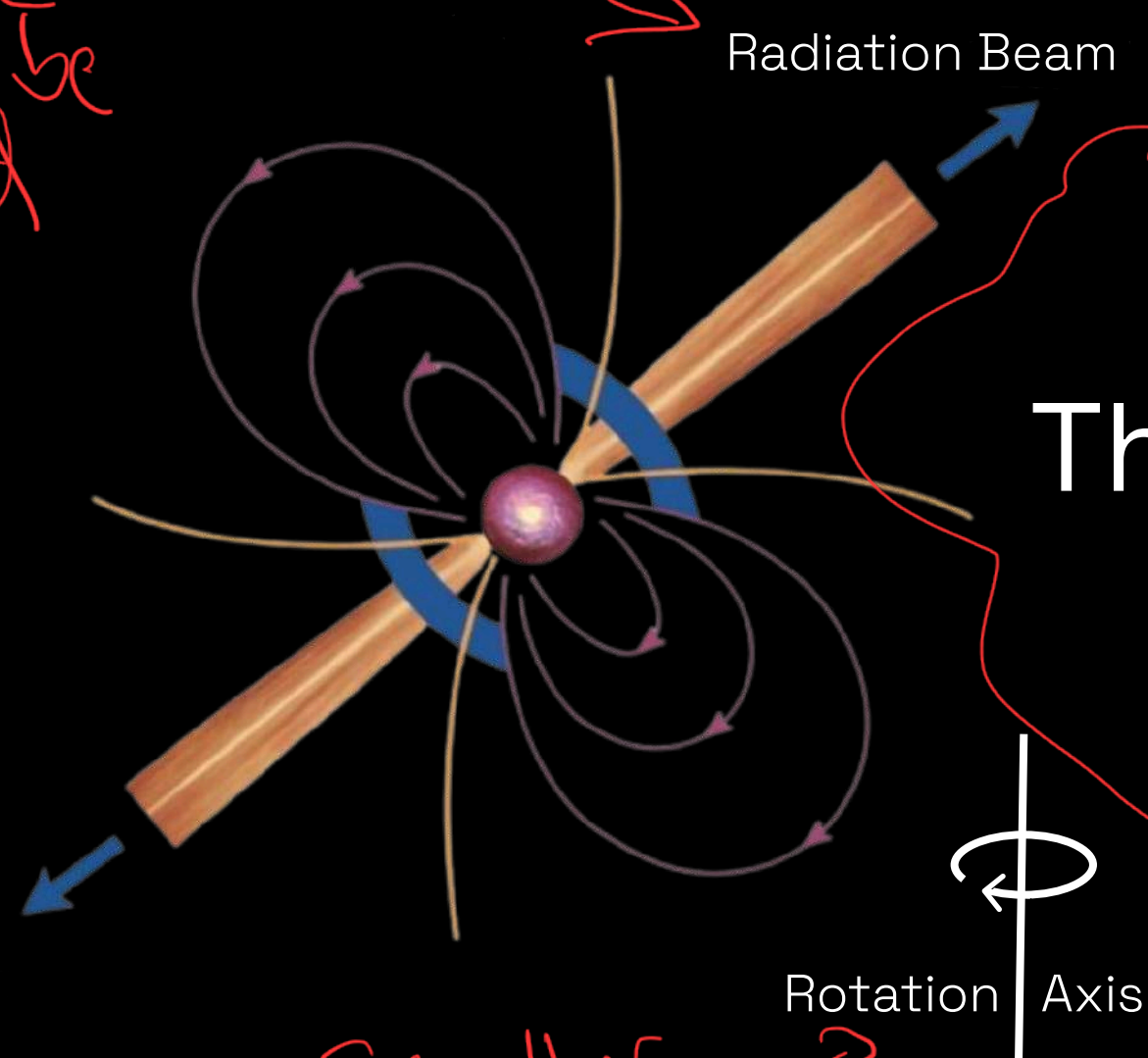
Pulsars are most easily found using radio waves, and can be used as clocks.

Explain  
mention  
radio  
telescopes

|                         |                   |
|-------------------------|-------------------|
| TYPE                    | NEUTRON STAR      |
| MASS                    | 1.4 SOLAR MASSES  |
| LIFESPAN                | 100 MILLION YEARS |
| RADIUS                  | 10-12 KM          |
| EXAMPLE OF NEUTRON STAR | VELA              |
| DISTANCE OF VELA TO SUN | ~1000 LIGHTYEARS  |

light

reference  
should be  
spelled  
out



smaller

**FUN FACT**

The fastest known pulsar spins at over 700 revolutions per second.



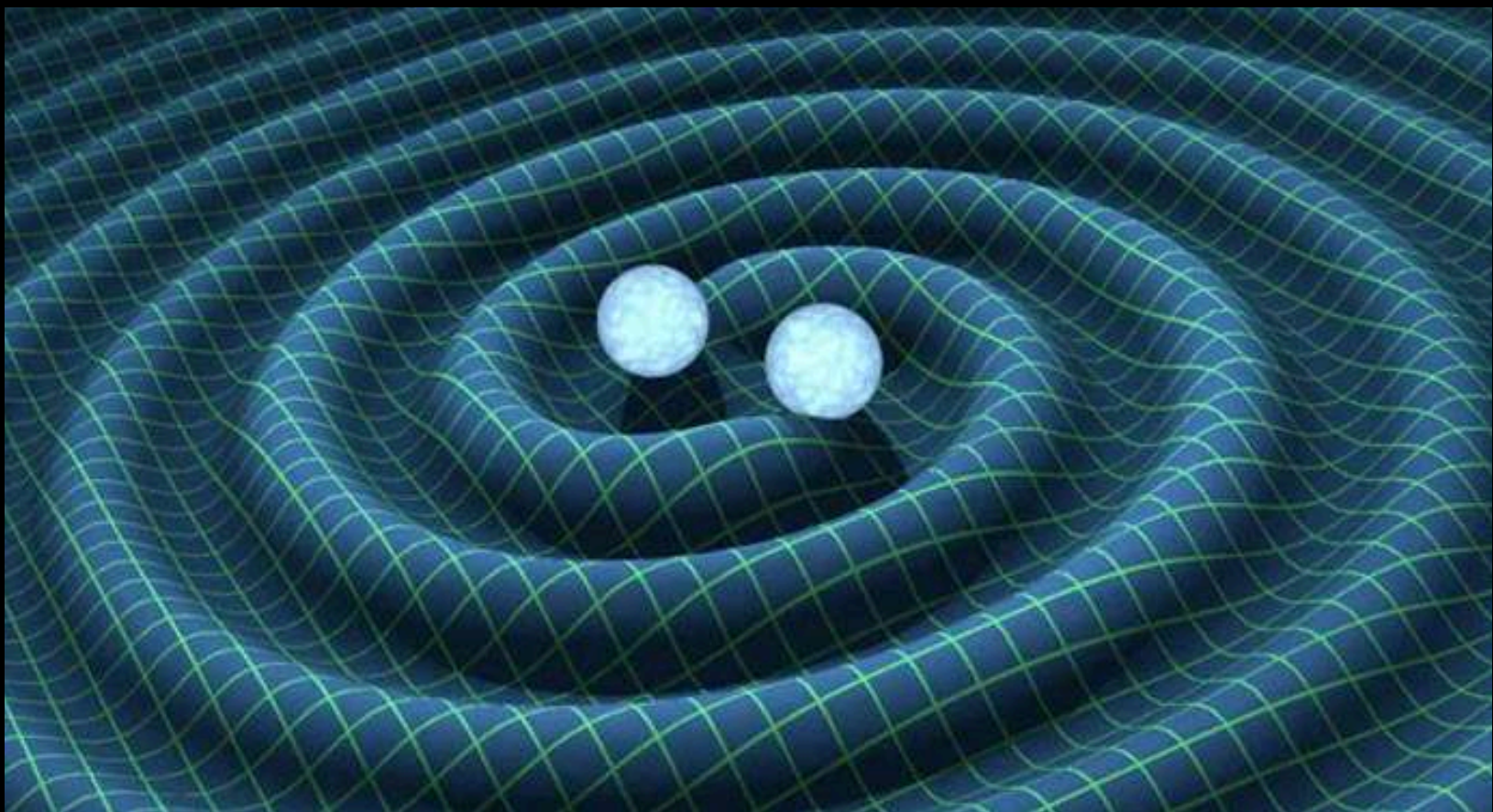
We cannot directly see black holes. Instead, we see stars and gas in an accretion disk fall into them. The disk is very hot from the friction of the matter, emitting x-rays that we can detect!



**Source:**

Event Horizon Telescope

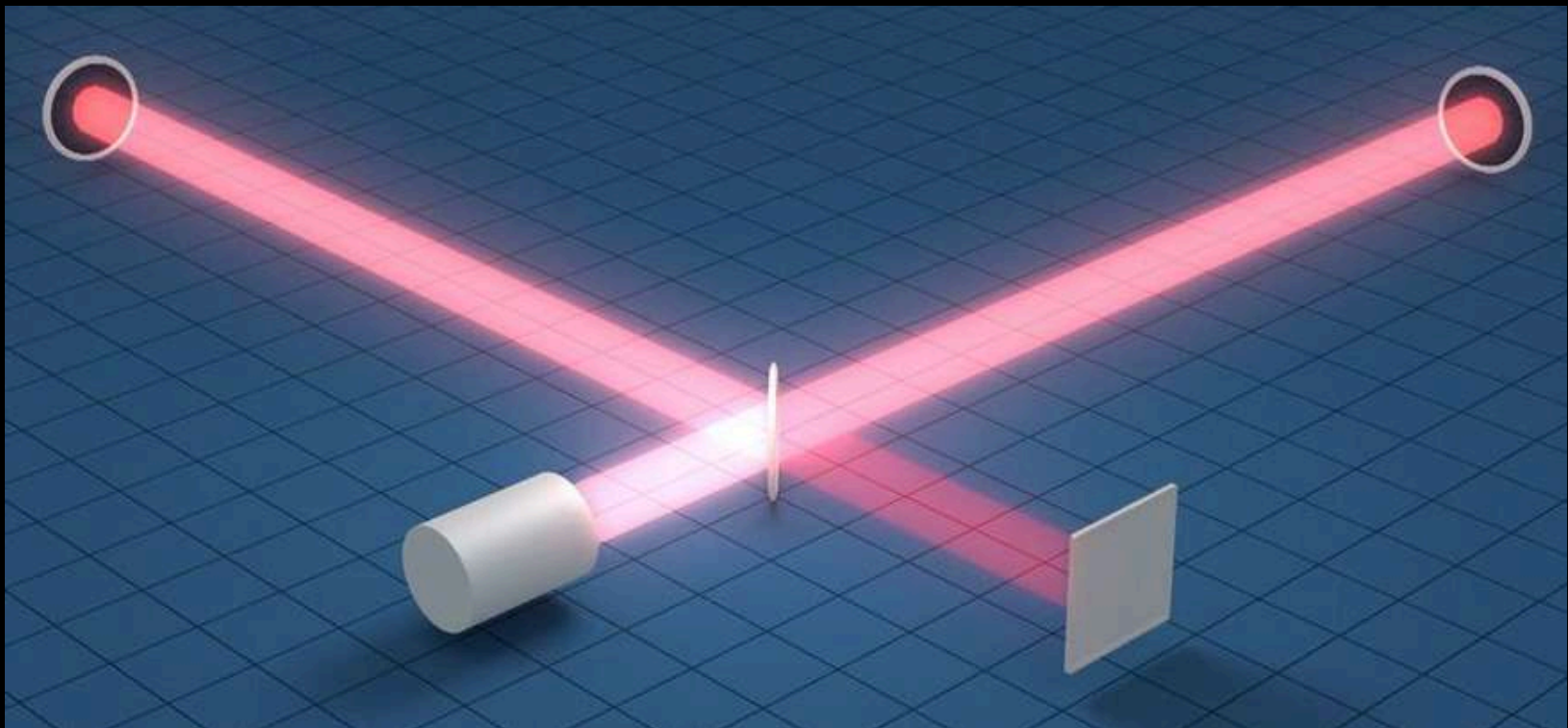
This is a photograph of the supermassive black hole at the center of the galaxy Messier 87. Here we can see a bright ring as light bends around it due to its immense gravity.



Visualisation of orbiting black holes creating gravitational waves

Source: R. Hurt/Caltech-JPL

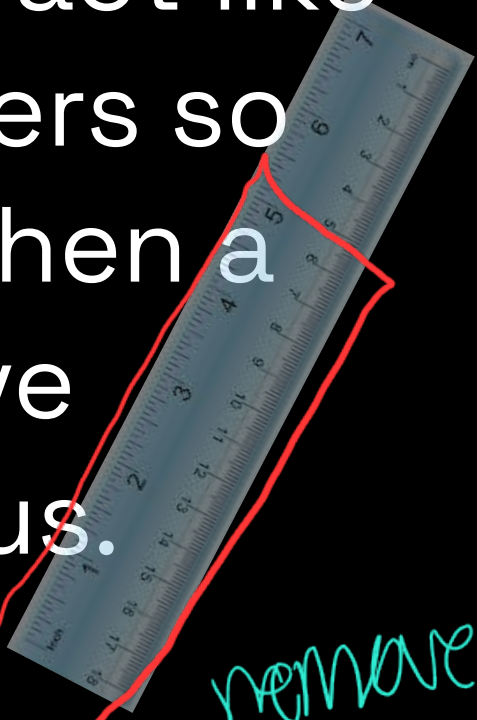
Black holes orbiting each other emit **gravitational waves** - ripples in the fabric of spacetime - which can be detected on Earth with special instruments.



Visualisation of how gravitational wave detectors work, checking for changes in length between mirrors.

Source: LIGO

Devices like LIGO act like really precise rulers so we can detect when a gravitational wave passes through us.







Stellar mass black holes are extremely heavy, but extremely small remains of supernovae. They have such strong gravity that even light can't escape!

Source: NASA's Goddard Space Flight Center/Jeremy Schnittman

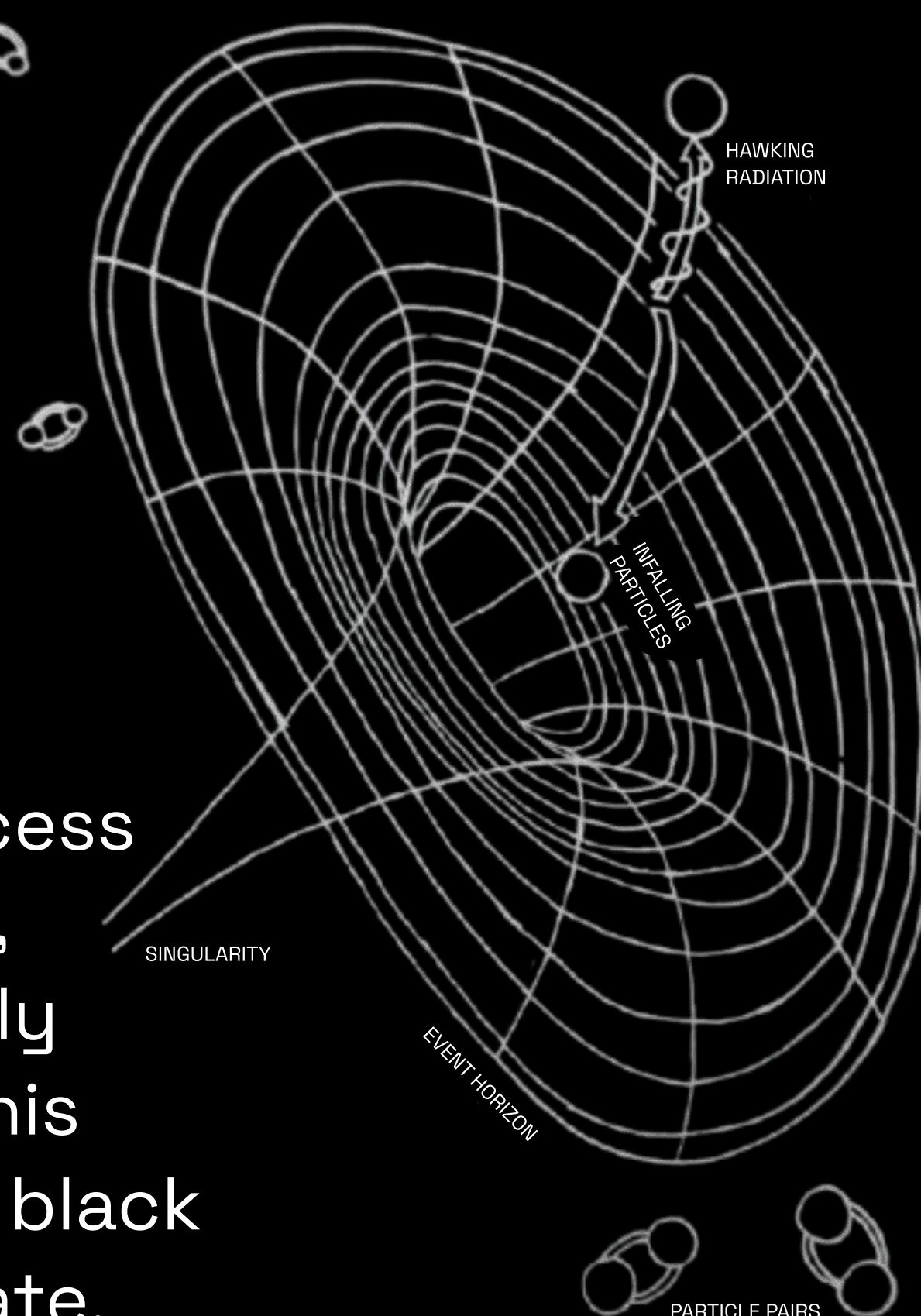
smaller to



Any object can become a black hole if it is compressed enough. The radius for this to happen is called the **Schwarzschild Radius**. For the Earth to be a black hole, you would have to squish it to be less than 2 cm across.

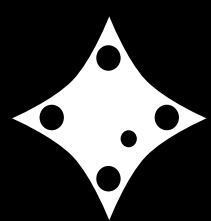
make text to the left  
instead of centered

Black holes are theorised to slowly decay through a process known as **Hawking radiation**, which works extremely slowly through quantum effects. This would mean that even giant black holes will eventually evaporate.

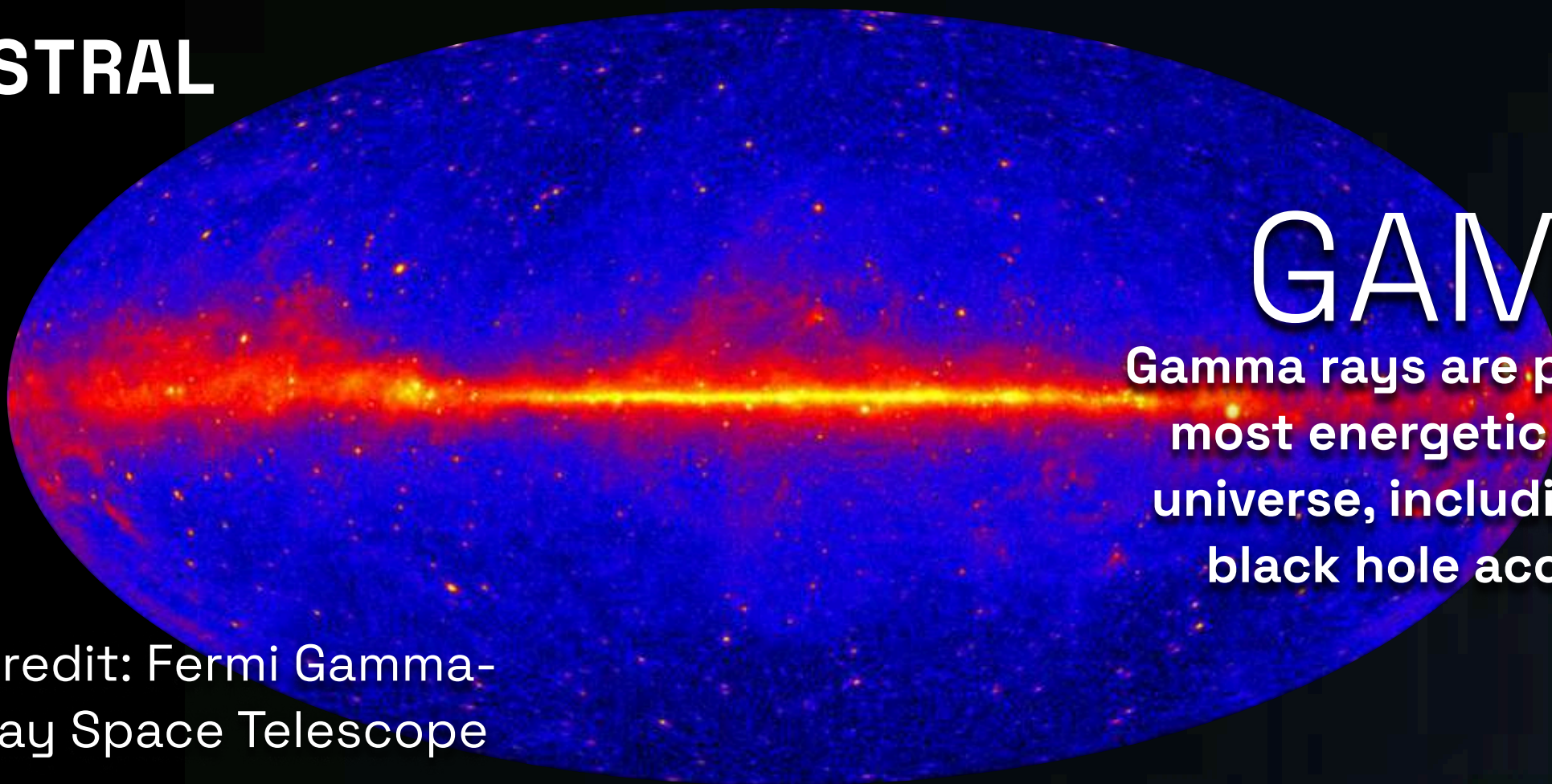


# BLACK HOLES





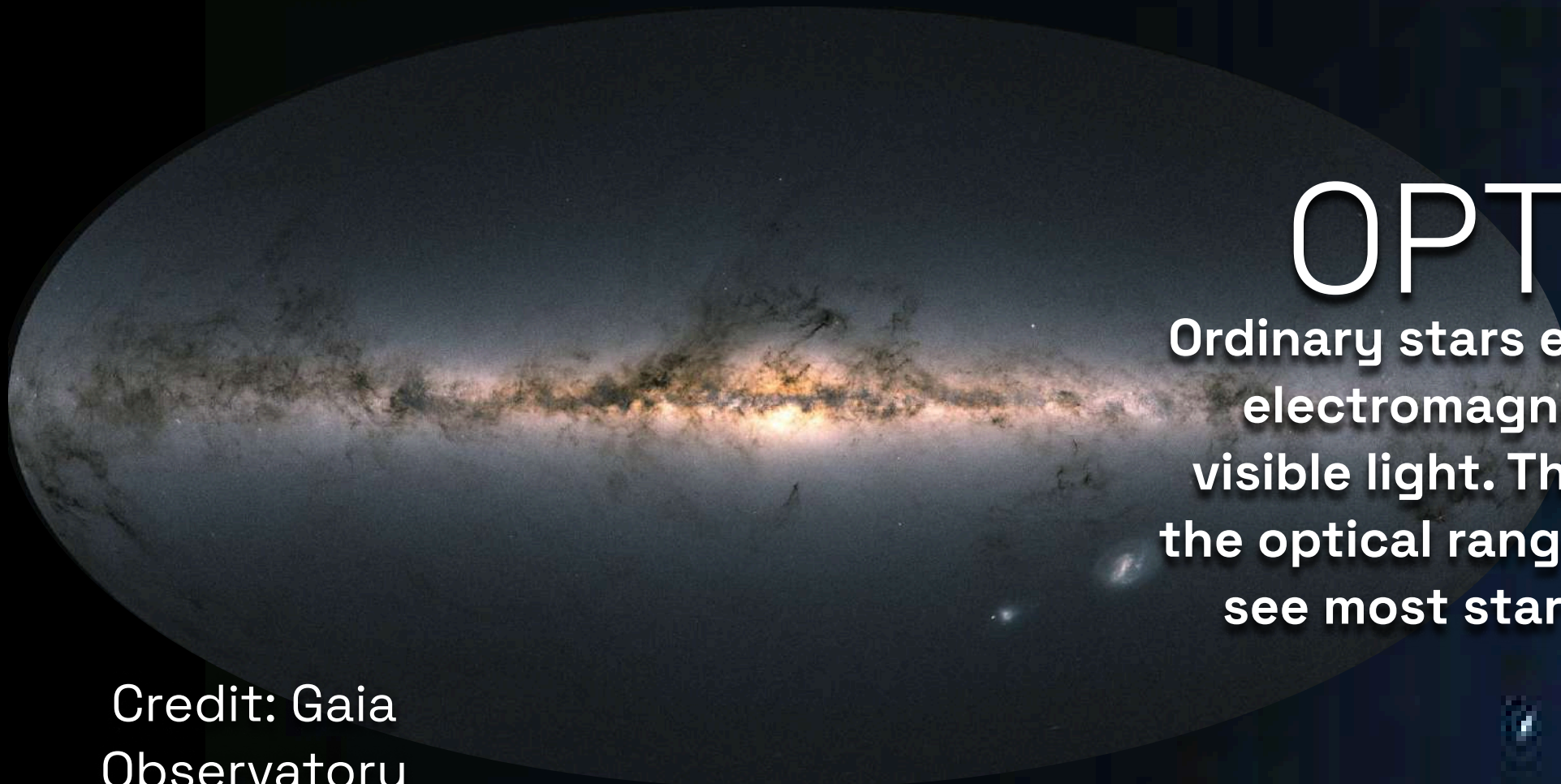
ASTRAL



## GAMMA

Gamma rays are produced by the most energetic objects in the universe, including pulsars and black hole accretion disks.

Credit: Fermi Gamma-ray Space Telescope



## OPTICAL

Ordinary stars emit most of their electromagnetic energy as visible light. This means that in the optical range, telescopes can see most stars and galaxies.

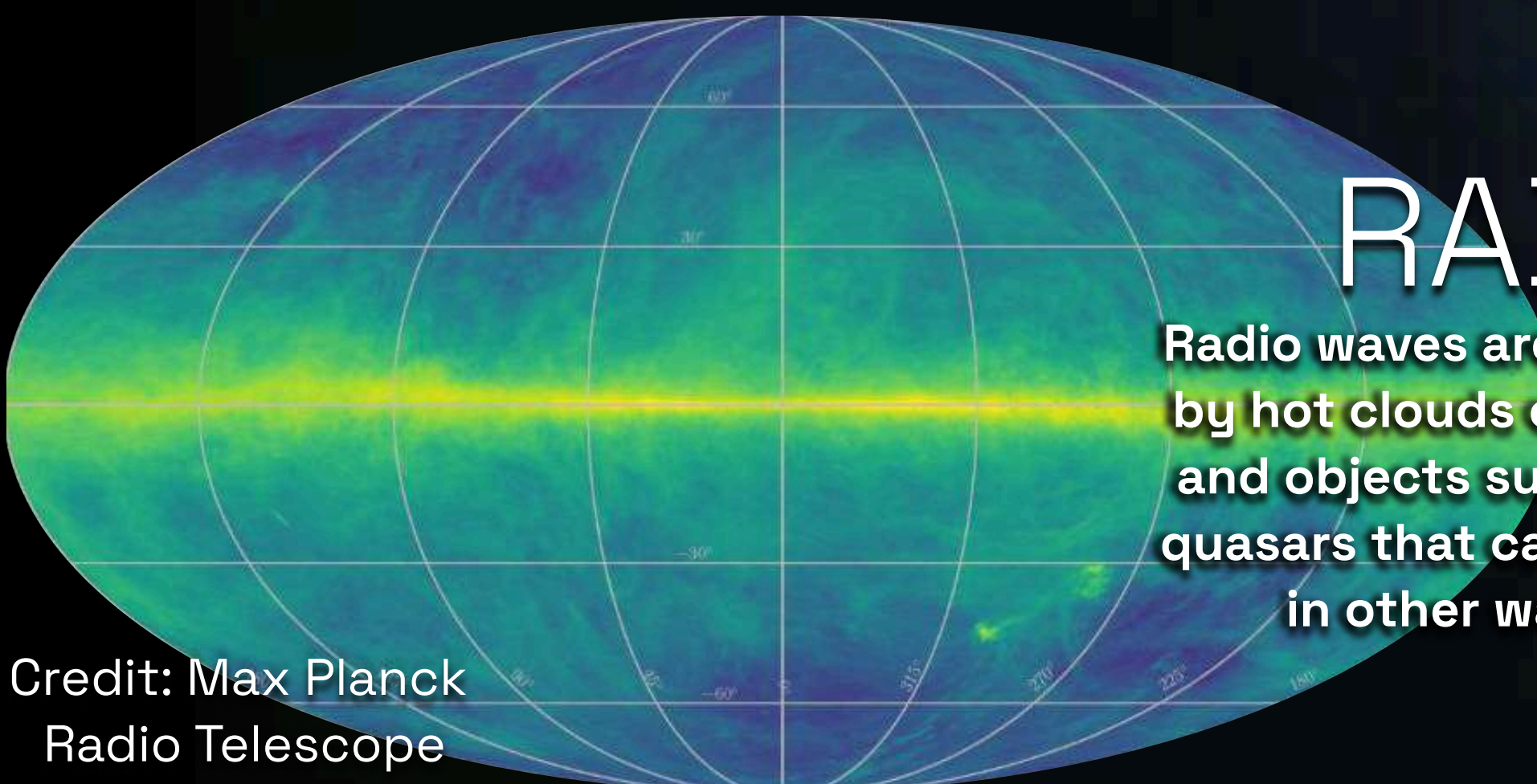
Credit: Gaia Observatory



## INFRARED

Infrared radiation is primarily emitted by stars and cooler objects such as exoplanets and clouds of dust in space.

Credit: Diffuse Infrared Background Experiment



## RADIO

Radio waves are mostly emitted by hot clouds of hydrogen gas and objects such as pulsars or quasars that can be hard to see in other wavelengths.

Credit: Max Planck Radio Telescope

1st 2nd 3rd 4th 5th 6th 7th 8th 9th 10th 11th 12th 13th 14th 15th 16th 17th 18th 19th 20th 21st 22nd 23rd 24th 25th 26th 27th 28th 29th 30th 31st 32nd 33rd 34th 35th 36th 37th 38th 39th 40th 41st 42nd 43rd 44th 45th 46th 47th 48th 49th 50th 51st 52nd 53rd 54th 55th 56th 57th 58th 59th 60th 61st 62nd 63rd 64th 65th 66th 67th 68th 69th 70th 71st 72nd 73rd 74th 75th 76th 77th 78th 79th 80th 81st 82nd 83rd 84th 85th 86th 87th 88th 89th 90th 91st 92nd 93rd 94th 95th 96th 97th 98th 99th 100th



vast empty space?



ASTRAL

ORDINARY  
MATTER

DARK  
MATTER

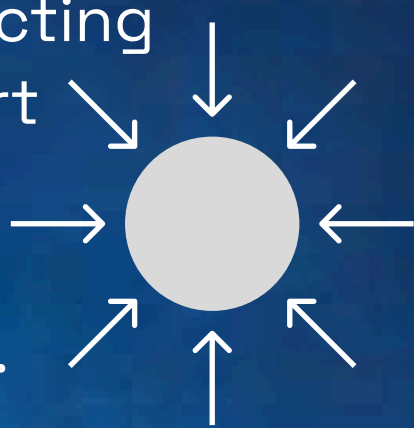
The Milky Way is believed to have around **10x** more mass than the stars we see.

Scientists call this invisible mass

# DARK MATTER.

## WIMPs

Some scientists think dark matter is made of 'Weakly Interacting Massive Particles' (WIMPs), which are heavy (so they exert a large gravitational force on other objects) but do not have a charge and are non-luminous.



## MACHOs

A popular theory used to be that dark matter was made of MACHOs (massively compact halo objects) that were the size of planets. These would be faint and very hard to detect, but eventually were ruled out as a dark matter candidate because they weren't found in large searches for them.

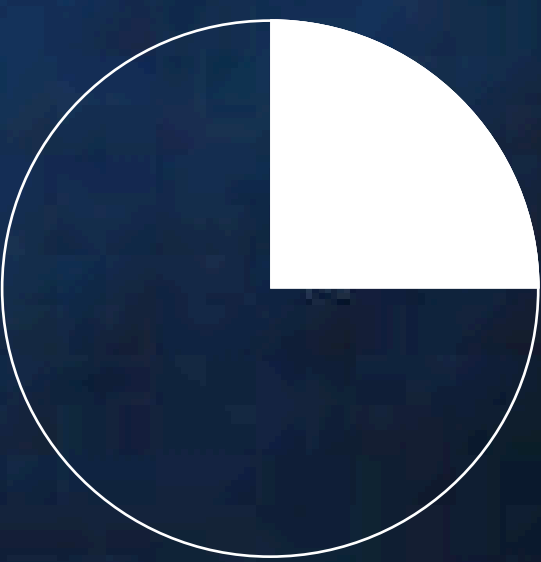


## MOND

MOND (Modified Newtonian Dynamics) theorises that Newton's laws can be modified to account for differences between the predicted and observed motion of galaxies. This theory could explain how galaxies and other astronomical objects move without dark matter.

The blue haze in this image represents the **DARK MATTER SURROUNDING** our galaxy.

Dark matter makes up around **25%** of our universe.

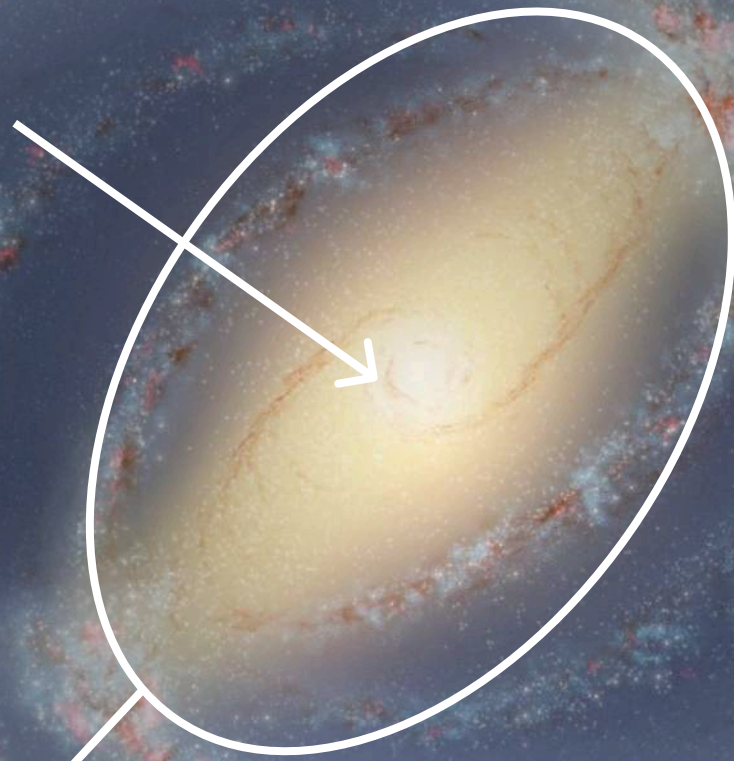


Background image credit:  
ESO/L. Calçada

THE MILKY WAY



There is a  
**SUPERMASSIVE BLACK  
HOLE** at the Milky Way's  
**GALACTIC  
CENTRE**



The Milky Way has **4**  
major **SPIRAL  
ARMS**

The **BAR** at the centre of the Milky Way  
is over **27,000**  
light-years across

The Milky Way's **DISC** is  
**9 BILLION** years old

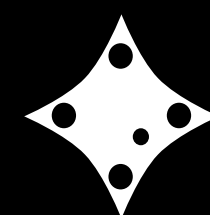
The **BULGE** is  
**10X** as thick as the  
disk of the galaxy

The stars and gases  
**SURROUNDING**  
the Milky Way are known as  
**THE HALO**

Source: 2MASS/JPL-Caltech



Source: NASA/Spitzer Telescope



ASTRAL

# GALAXIES

This galaxy is called IC 342,  
also known as “The Hidden Galaxy”  
because it is obscured behind large  
**CLOUDS OF DUST** in the Milky Way.

We can simulate galaxies  
like IC 342 on  
supercomputers, taking  
into account ordinary  
matter, gas, dark matter,  
dust clouds, star  
formation, and  
supernovae. The results  
can be very like real  
galaxies, as in the  
simulation shown here.

Source: FIRE Galaxy Simulations

space  
align



# Bibliography

<https://artsource-danielle.blogspot.com/2022/06/meerkat-telescopes-frb-artist-impression.html>

[https://www.uni-heidelberg.de/presse/meldungen/2019/m20190114\\_open-star-clusters-live-longer-than-previously-assumed.html](https://www.uni-heidelberg.de/presse/meldungen/2019/m20190114_open-star-clusters-live-longer-than-previously-assumed.html)

<https://starcycleitjseaberg.weebly.com/types-of-stars.html>

<https://www.space.com/22471-red-giant-stars.html>

[https://www.esa.int/Science\\_Exploration/Space\\_Science/Hubble\\_sees\\_faintest\\_stars\\_in\\_a\\_globular\\_cluster](https://www.esa.int/Science_Exploration/Space_Science/Hubble_sees_faintest_stars_in_a_globular_cluster)

<https://space-facts.com/galaxies/types/>

<https://www.nrao.edu/pr/2005/terzan5/>

<https://physicsworld.com/a/slowest-ever-neutron-star-is-found-in-cosmic-graveyard/>

<https://www.nationalgeographic.com/science/article/white-dwarfs>

[https://sites.ualberta.ca/~pogosyan/teaching/ASTRO\\_122/lect17/lecture17.html](https://sites.ualberta.ca/~pogosyan/teaching/ASTRO_122/lect17/lecture17.html)

**Scientific Notation**       $10^{10} = 10,000,000,000$   
                                  $10^7 = 10,000,000$

## Astronomical symbols

---

$L_{\odot}$       The luminosity of the Sun -  $3.8 \times 10^{26}$  watts

---

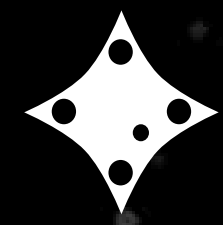
$M_{\odot}$       The mass of the Sun -  $1.989 \times 10^{30}$  kilograms

---

$R_{\odot}$       The radius of the Sun - 696,000 kilometres

---





# Glossary

## Terms

## Definitions

Accretion disk

A flattened, circular disk of gas and dust around a central object (such as a protostar, star or black hole) onto which matter is falling.

Dredge-up

A process which brings material to the surface of a star, from the regions deep within where nuclear fusion has occurred.

Globular Cluster

Compact systems of up to hundreds of thousand of stars bound together by gravity.

Ionization

The process of removing or adding electrons from atoms or molecules so that they carry a net electric charge.

Magnetic Winds

Ionised gas flowing outward from a star or other astronomical object, carrying along a magnetic field.

Protostar

A star in the process of formation. Protostars are often embedded deep in clouds of ionised gas and dust.

Quasar

A massive black hole in the central region of a galaxy, emitting bright radiation as matter falls on to it from an accretion disk.

+LIGD